MTH 205: ANALYSIS IN ONE VARIABLE

Lecture 1 DATE : 6th Jan 2025

- $\mathbb{R}$ – set of all real numbers
- For a point $x \in \mathbb{R}$ and $\varepsilon > 0$, the ~~ε-neighborhood~~ ε-neighborhood of x in $\mathbb{R}$ is given by the interval

$$N_\varepsilon(x) = (x-\varepsilon,\ x+\varepsilon)$$

- $\boxed{\text{NOTE}}$ the ε-neighborhood (in short we write "ε-nbd") is symmetric about the point x.

$\boxed{\text{Definition}}$ (open set)

A subset G of $\mathbb{R}$ is said to be <u>open set</u> if for each $x \in G$ there exists $\varepsilon_x > 0$ s.t $(x-\varepsilon_x, x+\varepsilon_x) \subseteq G$.

<u>Eg:</u>

① The set $\mathbb{R}$ is open.

For every $x \in \mathbb{R}$ choose $\varepsilon_x = 1$ we get

$$(x-1, x+1) \subset \mathbb{R}.$$

② $G = (0,1)$ is open.

Let $x \in G$. If $\underline{0 < x \leq 1/2}$ then

Choose $\varepsilon_x = x$ and so,

$$(x-\varepsilon_x, x+\varepsilon_x) \subset (0,1).$$

If $\underline{1/2 < x < 1}$ then choose $\varepsilon_x = 1-x$

and so,

$$(x-\varepsilon_x, x+\varepsilon_x) \subset (0,1).$$

③ Any open interval $I=(a,b)$ is an open set.

Hint: If $x \in I$ then choose

$$\varepsilon_x = \min\{x-a, b-x\}$$

and so that

$$(x-\varepsilon_x, x+\varepsilon_x) \subseteq I.$$

Similarly, show that $(-\infty, b)$ and (a,∞) are open sets

④ The empty set ϕ is open in $\mathbb{R}$

- The empty set ϕ contains no points at all, so ϕ is vacuously open set.

$\boxed{\text{Definition}}$ (closed set)

A subset F of $\mathbb{R}$ is said to be a closed set if $F^c = \mathbb{R} \setminus F$ is open set.

Equivalently, F is closed set if and only if for each $y \notin F$ there exists $\varepsilon_y > 0$ such that

$$F \cap (y - \varepsilon_y, y + \varepsilon_y) = \phi.$$

Eg: (1) The set $F = [0,1]$ is closed.

- Let $y \notin [0,1]$ then either $y < 0$ (or) $y > 1$.

If $y < 0$ then choose $\varepsilon_y = |y|$

and so, $(y-\varepsilon_y, y+\varepsilon_y) \cap [0,1] = \phi$.

If $y>1$ then choose $\varepsilon_y = y-1$

and so we have

$$(y-\varepsilon_y, y+\varepsilon_y) \cap [0,1] = \phi.$$

2 The set $\{x \in \mathbb{R} / 0 \le x < 1\}$ is neither open nor closed.

- The given set is $[0,1)$ clearly.

$\forall \varepsilon > 0$,

$$(-\varepsilon, \varepsilon) \not\subset [0,1)$$

- Hence $[0,1)$ is NOT open.
- Similarly, $1 \notin [0,1)$ but for any $r>0$ the interval

$(1-r, 1+r) \cap [0,1) \neq \phi.$

$\Rightarrow$ $[0,1)$ is NOT closed in $\mathbb{R}$.

(3) $\mathbb{R}$ is both closed and open.

- Clearly, $\mathbb{R}$ is open. Since the empty set $\phi = \mathbb{R} \setminus \mathbb{R}$ is open, $\mathbb{R}$ is closed.

(4) Similarly, ϕ is both closed and open.

(5) Infact, every closed interval $[a,b]$ is a closed set. (Exercise).

$\boxed{\text{PROPERTIES}}$:- Open Sets :-

① The union of arbitrary collection of open subsets in $\mathbb{R}$ is open.

Let $\{G_\alpha : \alpha \in \Lambda\}$ be a family of open subsets in $\mathbb{R}$. We claim that the union $G = \bigcup_{\alpha \in \Lambda} G_\alpha$ is open.

Consider $x \in G$, there is some $\beta \in \Lambda$ with $x \in G_\beta$. Since G_β is an open set $\exists\ \varepsilon_x > 0$ such that

$$(x-\varepsilon_x, x+\varepsilon_x) \subseteq G_\beta \subseteq \bigcup_{\alpha \in \Lambda} G_\alpha$$

$$\Rightarrow (x-\varepsilon_x, x+\varepsilon_x) \subseteq G.$$

Since x is an arbitrary element of G, we conclude that G is open $\square$

② The intersection of any finite collection of open sets in $\mathbb{R}$ is open.

Suppose $G_1, G_2, G_3, \ldots G_n$ are open sets in $\mathbb{R}$. We claim that $G := \bigcap_{j=1}^{n} G_j$ is open.

If $x \in G$, then $x \in G_j$, $1 \le j \le n$.
Since each G_j is open $\exists\ \varepsilon_j > 0$
s.t. $(x - \varepsilon_j, x + \varepsilon_j) \subseteq G_j$, $1 \le j \le n$

By choosing $\varepsilon = \min\{\varepsilon_1, \varepsilon_2, \ldots \varepsilon_n\}$ we see that

$$(x - \varepsilon, x + \varepsilon) \subseteq G.$$

Since x was chosen arbitrarily, we conclude that G is open.

Remark The intersection of infinitely many open sets in $\mathbb{R}$ need not be open.

Eg: Let $G_n := (0, 1+\frac{1}{n})$ for $n \in \mathbb{N}$.

Clearly, G_n is open for each n.

However, $\bigcap_{n=1}^{\infty} G_n = (0, 1]$ which

is NOT open.

③ The intersection of an arbitrary collection of closed sets in $\mathbb{R}$ is closed.

Suppose $\{F_\alpha : \alpha \in \Lambda\}$ is a family of closed sets in $\mathbb{R}$.

We claim that $F = \bigcap_{\alpha \in \Lambda} F_\alpha$ is closed

By De Morgan identity, we have

$$F^c = \bigcup_{\alpha \in \Lambda} F_\alpha^c .$$

Since each F_α^c is open, it follows from property 1 that F^c is open.

Consequently, F is closed.

4 The union of any finite collection of closed sets in $\mathbb{R}$ is closed.

Suppose $F_1, F_2, \ldots, F_n$ are closed sets in $\mathbb{R}$ and let $F = \bigcup_{j=1}^{n} F_j$. Since each F_j^c is open, by property 2 we see that $F^c = \bigcap_{j=1}^{n} F_j^c$ is open

Hence F is closed.

Remark

The union of infinitely many closed sets in $\mathbb{R}$ need not be closed.

eg: Let $F_n = \left[\frac{1}{n}, 1\right]$ for $n \in \mathbb{N}$.

Here each F_n is closed.

However, the union

$$F = \bigcup_{n=1}^{\infty} F_n = (0, 1] ,$$

which is not closed.

Characterizations

Theorem [10]: Let $F \subseteq \mathbb{R}$ then the following are equivalent.

(i) F is a closed subset of $\mathbb{R}$.

(ii) If $\{x_n\}$ is any convergent sequence of elements in F then $\lim_{n\to\infty} x_n \in F$.

Proof.

(i) $\Rightarrow$ (ii): Given that F is closed.

Assume that $\{x_n\}$ sequence in F such that $x_n \longrightarrow x$ as $n \to \infty$ but $x \notin F$.

This means that $x \in F^c$, which is open

So $\exists\ \varepsilon > 0$ s.t

$$(x-\varepsilon\ ,\ x+\varepsilon) \subseteq F^c.$$

Since $\{x_n\} \longrightarrow x$ there is $N_\varepsilon \in \mathbb{N}$

with

$$x_n \in (x-\varepsilon\ ,\ x+\varepsilon) \subseteq F^c$$

for $n \geqslant N_\varepsilon$

This is a Contradiction as $\{x_n\}$ is in F.

Therefore, we get $x \in F$.

(ii) $\Rightarrow$ (i):

Suppose, on the contrary, that F is NOT closed.

Equivalently, F^c is NOT open.

There exist $y \in F^c$ such that for each $n \in \mathbb{N}$ we have

$$\left(y - \frac{1}{n}, y + \frac{1}{n}\right) \cap F \neq \phi$$

Let $y_n \in \left(y - \frac{1}{n}, y + \frac{1}{n}\right) \cap F$, $\forall n \in \mathbb{N}$

Then $\{y_n\}$ is a sequence in F

and $|y_n - y| < \frac{1}{n}$

$\Rightarrow \quad |y_n - y| \longrightarrow 0$ as $n \to \infty$

We have shown that

F is NOT closed

$\Rightarrow$ (ii) is NOT true.

i.e., $\sim(\text{i}) \Rightarrow \sim(\text{ii})$

Consequently,

(ii) $\Rightarrow$ (i) holds true.

$\boxed{\text{Definition}}$ (cluster point)

Let S be a subset of $\mathbb{R}$.

A point $x \in \mathbb{R}$ is called

a cluster point (or) limit point

of S in $\mathbb{R}$ if

"every ε-nbd of x contains

a point of F different from x"

i.e.,

$$\left[F \cap (x-\varepsilon, x+\varepsilon)\right] \setminus \{x\} \neq \phi,$$

$\forall\, \varepsilon > 0.$

Remark: A point $x \in \mathbb{R}$ is a cluster point of S in $\mathbb{R}$ if and only if $x = \lim_{n\to\infty} x_n$ for some sequence $\{x_n\}$ in S.

Suppose $x \in \mathbb{R}$ is a cluster point of S in $\mathbb{R}$. For each $n \in \mathbb{N}$, one can choose $x_n \neq x$ s.t

$$x_n \in \left(x - \frac{1}{n}, x + \frac{1}{n}\right) \cap S.$$

$$\Rightarrow \quad |x_n - x| < \frac{1}{n} \longrightarrow 0 \text{ as } n \to \infty$$

$$\Rightarrow \quad x = \lim_{n\to\infty} x_n, \text{ for } \{x_n\} \text{ in } S.$$

Conversely, if $x = \lim_{n\to\infty} x_n$ for some sequence $\{x_n\}$ in S then x is a cluster point of "S" in $\mathbb{R}$

$\boxed{\text{Theorem}}$ A subset F of $\mathbb{R}$ is closed if and only if it contains all of its cluster points.

$\boxed{\text{EX}}$ Show that every open set in $\mathbb{R}$ can be written as a union of countably many disjoint open intervals in $\mathbb{R}$.

Is it possible to express every closed subset of $\mathbb{R}$ as a countable intersection of closed intervals?

The answer is No (why?)

$\boxed{\text{The Cantor set}}$

- Consider the closed interval $[0,1]$

- Remove the middle third $\left(\frac{1}{3}, \frac{2}{3}\right)$ of $[0,1]$, we obtain

$$C_1 := \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right]$$

- Now remove the open middle third of each of the two closed intervals in C_1, we get

$$C_2 := \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right]$$

- Here C_2 is the union of 2^2 closed intervals.

- By continuing this process, we obtain C_3 as the union of 2^3 closed intervals of the form $\left[\frac{k}{3^3}, \frac{k+1}{3^3}\right]$

- In this process we get a sequence $\{C_n\}_{n=1}^{\infty}$, where C_n is the union of 2^n closed intervals of the form

$$\left[\frac{k}{3^n}, \frac{k+1}{3^n}\right]$$

- The "Cantor set" is defined by

$$C = \bigcap_{n=1}^{\infty} C_n .$$

Properties

1. The Sum of the lengths of the removed intervals is 1.

- In the first step we remove an interval of length $\frac{1}{3}$.
- In the second step we remove two intervals of each having length $\frac{1}{3^2}$
  The total length is, $\frac{2}{3^2}$
- In the third step we remove 2^2 intervals of each having length $\frac{1}{3^3}$
  The total length is, $\frac{2^2}{3^3}$.
- In general, at the n^{th} step, the total length is, $\frac{2^{n-1}}{3^n}$.

- The total length L of the removed intervals is given by

$$L = \frac{1}{3} + \frac{2}{3^2} + \frac{2^3}{3^3} + \frac{2^4}{3^4} + \cdots$$

$$= \frac{1}{3}\left[1 + \frac{2}{3} + \frac{2^3}{3^3} + \cdots + \frac{2^n}{3^n} + \cdots\right]$$

$$= \frac{1}{3}\sum_{n=0}^{\infty}\left(\frac{2}{3}\right)^n$$

$$= \frac{1}{3}\cdot\frac{1}{1-\frac{2}{3}}$$

$$= \frac{1}{3}\cdot 3$$

$$= 1$$

② The Cantor set 'C' Contains no non-empty interval as a subset.

Suppose 'C' Contains a non-empty open interval $I = (a,b)$.

That is, $(a,b) \subseteq C_n$ for each $n \in \mathbb{N}$.

Since C_n is the union of 2^n closed intervals, each of length $\frac{1}{3^n}$, the total length of the intervals is, $\left(\frac{2}{3}\right)^n$

$\Rightarrow \quad 0 < \text{length of } (a,b) \leq \left(\frac{2}{3}\right)^n \quad \forall n \in \mathbb{N}$

$\Rightarrow \quad 0 < b-a < \left(\frac{2}{3}\right)^n, \ \forall n \in \mathbb{N}$

Here $\left(\frac{2}{3}\right)^n \longrightarrow 0$ as $n \to \infty$.

$\Rightarrow \quad b = a.$

Then I is empty, this is a contradiction.

Therefore, the Cantor set contains no non-empty open intervals.

③ The Cantor set 'C' is uncountable

Firstly note that any $x \in [0,1]$ can be written in a ternary (base 3) expansion.

$$x = \sum_{n=1}^{\infty} \frac{a_n}{3} = (0.a_1 a_2 a_3 \cdots)_3$$

where $a_n \in \{0,1,2\}$

For instance each point in $\left(\frac{1}{3}, \frac{2}{3}\right)$

has $a_1 = 1$.

The end points of the removed intervals have two possible ternary expansion: one expansion having 1's ; one having no 1's.

For example;

$$\frac{1}{3} = (0.10000\cdots)_3$$

&

$$\frac{1}{3} = (0.02222222\cdots)_3$$

Now we identify 'C' with all $x \in [0,1]$ thave ternary expansion with no 1's.

Define

$$f : C \longrightarrow [0,1] \text{ by}$$

$$f\left(x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}\right) := \sum_{n=1}^{\infty} \frac{(a_n/2)}{2^n}.$$

$$\forall x \in C.$$

That is,

$$f\left((0.a_1a_2\cdots)_3\right)$$

$$= (0.b_1b_2b_3\cdots)_2$$

where $b_n = \frac{a_n}{2}$ $\forall n \in \mathbb{N}$

Thus f is a surjective map.

(why?)

$\therefore$ C is uncountable set.

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Lecture 3 DATE : 13th Jan 2025

COMPACT SETS

Definition: Let $A \subseteq \mathbb{R}$

An open cover of A is a collection $G = \{G_\alpha\}$ of open subsets of $\mathbb{R}$ with $A \subseteq \bigcup_\alpha G_\alpha$.

- If G' is a subcollection of sets from G such that the union of sets in G' also contains A then G' is called a subcover of G.
- If G' consists of finitely many sets then we call G' a finite subcover of G.

Examples:

① Let $A = [1, \infty)$

- $\{(0, \infty)\}$ – open cover of A

- $\{(n-1, n+1) : n \in \mathbb{N}\}$

  – open cover of A

- $\{(0, n) : n \in \mathbb{N}\}$ – open cover of A

- $\{(r-1, r+1) : r \in \mathbb{Q}, r > 0\}$

  – open cover of A

There can be many different open covers for a given set.

Definition A subset K of $\mathbb{R}$ is said to be compact if every open cover of K has a finite subcover.

"In otherwords, K is compact, if whenever it is contained in the union of a collection $\mathcal{G} = \{G_\alpha\}$ of open sets in $\mathbb{R}$ then $\exists$ a finite subcollection $\{G_1, G_2, \cdots G_n\}$ of $\mathcal{G}$ such that

$$K \subseteq \bigcup_{i=1}^{n} G_i.$$

Examples:

(1) Let $K = \{x_1, x_2, x_3, x_4, \ldots x_n\}$ be a finite subset of $\mathbb{R}$.

If $G = \{G_\alpha\}_{\alpha \in \Lambda}$ is an open cover of the set K then for each x_i $\exists$ G_{α_i} s.t

$$x_i \in G_{\alpha_i} \qquad i = 1, 2, 3, \ldots n$$

and so,

$$\{G_{\alpha_1}, G_{\alpha_2}, \ldots, G_{\alpha_n}\}$$

is a finite subcover of G.

$\therefore$ K is Compact.

- Every finite subset of $\mathbb{R}$ is compact.

(2) $S = [0, \infty)$, this is NOT compact.

Let $G_n = (-1, \infty)$ for each $n \in \mathbb{N}$

then
$$S \subseteq \bigcup_{n=1}^{\infty} G_n$$

This means $\mathcal{G} = \{G_n : n \in \mathbb{N}\}$ is an open cover of S.

Suppose that $\{G_{n_1}, G_{n_2}, \dots G_{n_\ell}\}$ is any finite subcollection of $\mathcal{G}$. If we take

$$m := \sup\{n_1, n_2, \dots, n_\ell\}$$

then

$$G_{n_1} \cup G_{n_1} \cup \dots \cup G_{n_\ell}$$
$$= G_m$$
$$= (-1, m).$$

Clearly, this union does NOT contain S.

Therefore, S has an open cover that has no finite subcover.

Hence S is NOT compact.

(3) $S = (0,1)$ is NOT Compact

By taking $G_n = \left(\frac{1}{n}, 1\right)$ $\forall n \in \mathbb{N}$

then
$$(0,1) = \bigcup_{n=1}^{\infty} \left(\frac{1}{n}, 1\right)$$

and so,

$\mathcal{G} = \{ G_n : n \in \mathbb{N} \}$ is an open cover of S.

If $\{ G_{n_1}, G_{n_2}, \ldots, G_{n_t} \}$ is any finite sub collection of $\mathcal{G}$, then

$$G_{n_1} \cup G_{n_2} \cup \ldots \cup G_{n_t} = \left(\frac{1}{m}, 1\right)$$

where
$$m := \sup \{ n_1, n_2, \ldots n_t \}$$

This union does NOT contain S.

$\therefore$ $S = (0,1)$ is NOT compact.

Now we identify all compact subsets of $\mathbb{R}$

<u>Theorem [102]</u>:

If K is a compact subset of $\mathbb{R}$ then K is closed and bounded.

Proof. Firstly, we show that K is bounded.

let $G_n = (-n, n)$ for each $n \in \mathbb{N}$

then $\{G_n : n \in \mathbb{N}\}$ is an open cover of K since

$$K \subseteq \mathbb{R} = \bigcup_{n=1}^{\infty} G_n.$$

Since K is compact, there is a finite subcover i.e., $\exists\, m \in \mathbb{N}$ s.t

$$K \subseteq G_m = (-m, m).$$

Therefore, K is bounded.

Now we show that K is closed. It is enough to prove that K^c is open.

let $x \in K^c$, for each $n \in \mathbb{N}$

take

$$S_n = \left\{ y \in \mathbb{R} \;\middle/\; |y-x| > \frac{1}{n} \right\}$$

then each S_n is open (why?)

with

$$\mathbb{R} \setminus \{x\} = \bigcup_{n=1}^{\infty} S_n .$$

and

$$K \subseteq \bigcup_{n=1}^{\infty} S_n .$$

Since K is compact there

is a subcollection

$\{ S_{n_1}, S_{n_2}, \ldots, S_{n_\ell} \}$ s.t

$$K \subseteq \bigcup_{i=1}^{\ell} S_{n_i}$$

By taking $m = \sup\{n_1, n_2, \ldots n_\ell\}$

we have

$$K \subseteq S_m = \mathbb{R} \setminus \left(x - \tfrac{1}{m}, x + \tfrac{1}{m}\right)$$

It follows that

$$K \cap \left(x - \tfrac{1}{m}, x + \tfrac{1}{m}\right) = \phi.$$

Equivalently,

$$\left(x - \tfrac{1}{m}, x + \tfrac{1}{m}\right) \subseteq K^c.$$

Thus K^c is open and hence K is closed.

Theorem [103] (Heine-Borel theorem.)

A subset K of $\mathbb{R}$ is Compact if and only if K is closed and bounded.

Proof. We know from Theorem [102] that a Compact subset of $\mathbb{R}$ is closed and bounded.

Conversely, suppose that K is closed and bounded. Let $\mathcal{G} = \{G_\alpha\}_{\alpha\in\Lambda}$ be an open cover of K.

Assume that K has no finite subcover of $\mathcal{G}$

From the hypothesis, K is bounded so there exists $r > 0$ such that

$$K \subseteq [-r, r].$$

We denote by $I_1 = [-r, r]$

$$I_1' = [-r, 0], \quad I_1'' = [0, r]$$

Atleast one of the two subsets $K \cap I_1'$ and $K \cap I_1''$ is NOT contained in the union of any finite subcollection of $\mathcal{G}$.

If $K \cap I_1'$ is not contained in the union of any finite subcollection of $\mathcal{G}$, we take

$$I_2 := I_1'$$

Now we bisect I_2 into two closed subintervals I_2' & I_2''.

If $K \cap I_2'$ is not contained in the union of any finite subcollection of $\mathcal{G}$, we take

$$I_3 := I_2'$$

Continuing this process we obtain a nested sequence of intervals $\{I_n\}$ with

$I_1 \supset I_2 \supset I_3 \supset \cdots$

$\Rightarrow \bigcap_{n=1}^{\infty} I_n \neq \phi$

$\exists \; z \in \bigcap_{n=1}^{\infty} I_n$

(show that) z is a cluster point of K.

Since K is closed, $z \in K$.

There exists a set G_λ in $\mathcal{G}$ with $z \in G_\lambda$. As G_λ is open

$\exists \; \varepsilon > 0$ s.t

$(z-\varepsilon, z+\varepsilon) \subseteq G_\lambda$.

The length of interval I_n is, $\frac{r}{2^{n-2}}$, $\forall n \in \mathbb{N}$.

$\exists N \in \mathbb{N}$ s.t

$$\frac{r}{2^{N-2}} < \varepsilon$$

$$\Rightarrow I_N \subseteq (z-\varepsilon, z+\varepsilon) \subseteq G_\lambda.$$

This means that

$$K \cap I_N \subseteq G_\lambda$$

which is a contradiction to our construction of I_N.

Our assumption is wrong.

There is a finite subcover of K.

Hence K is compact. $\square$

Now we combine the Heine-Borel theorem with Bolzano-Weierstrass theorem to obtain a sequential characterization of compact subsets of $\mathbb{R}$

<u>Theorem [104]</u> A subset K of $\mathbb{R}$ is compact if and only if every sequence in K has a subsequence that converges to a point in K.

Proof. Suppose K is compact and let $\{x_n\}$ be a sequence in K. By the Heine-Borel theorem, the set K is bounded and so, $\{x_n\}$ is bounded.

From Bolzano-Weierstrass theorem there is a subsequence $\{x_{n_k}\}$ of $\{x_n\}$ such that

$$x_{n_k} \longrightarrow x \text{ (say)}, \text{ as } n \to \infty$$

Since K is closed, $x \in K$.

To prove the Converse, we will show that if K is either NOT closed or NOT bounded, then there must exist a sequence in K that has <u>no</u> converging subsequence.

<u>If K is NOT closed</u>, there is a cluster point 'x' of K s.t $x \notin K$. There is a sequence $\{x_n\}$ in K s.t

$$x_n \longrightarrow x \quad \text{as } n \to \infty$$

Here $x_n \neq x$ for any $n \in \mathbb{N}$

and $x \notin K$, so there does NOT exist any subsequence of $\{x_n\}$ that converges to a point of K.

<u>If K is NOT bounded</u>, then there exists a sequence $\{x_n\}$ in K with $|x_n| > n$ for all $n \in \mathbb{N}$.

Then every subsequence of $\{x_n\}$ is unbounded and so, it can NOT converge (indeed to any point of K).

Hence the result.

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Lecture 3 DATE : 20.01.2025

Continuous Function

Let $A \subseteq \mathbb{R}$, let $f : A \longrightarrow \mathbb{R}$ be a function and $a \in A$.

We say that f is continuous at a if

$\forall\ \varepsilon > 0\ \exists\ \delta > 0$ such that

$|f(x) - f(a)| < \varepsilon$ whenever $|x - a| < \delta$.

In other words, $\forall\ \varepsilon > 0\ \exists\ \delta > 0$ s.t

$x \in (a - \delta, a + \delta)$

$\implies f(x) \in (f(a) - \varepsilon, f(a) + \varepsilon)$

Definition : A function $f : A \to \mathbb{R}$ is said to be continuous at $a \in A$ if for every $\varepsilon > 0$ $\exists$ $\delta > 0$ such that

$$f\left(N_\delta(a)\right) \subseteq N_\varepsilon\left(f(a)\right).$$

- If f is continuous at every $a \in A$ then f is called "continuous" on A.
- If 'f' fails to be continuous at $a \in A$ then we say that f is <u>discontinuous</u> at a.

Theorem [105]

Let $A \subseteq \mathbb{R}$ and $f: A \longrightarrow \mathbb{R}$ be a function. Then the following are equivalent:

① f is continuous

② $\{f(x_n)\} \longrightarrow f(x)$ whenever $\{x_n\} \longrightarrow x$ in A.

③ $f(\overline{B}) \subset \overline{f(B)}$ for every subset $B \subset A$.

④ $f^{-1}(Z)$ is closed in A for every closed set $Z \subset \mathbb{R}$.

⑤ $f^{-1}(V)$ is open in A for every open set $V \subset \mathbb{R}$.

Proof.

①$\Rightarrow$②: Let $\{x_n\} \longrightarrow x$ in A and $\varepsilon > 0$.

We show that $f(x_n) \longrightarrow f(x)$.

Since f is continuous, there exists $\delta > 0$ such that

$$f(N_\delta(a)) \subset N_\varepsilon(f(a)), \quad \forall\, a \in A.$$

As $\{x_n\} \longrightarrow x$ there is $N \in \mathbb{N}$ s.t. $x_n \in N_\delta(x)$ $\forall$ $n \geq N$.

Then $f(x_n) \in N_\varepsilon(f(x))$

$\forall\, n \geq N.$

i.e., $|f(x_n) - f(x)| < \varepsilon$

$\forall\, n \geq N.$

It follows that

$\{f(x_n)\} \longrightarrow f(x)$

as $n \to \infty$.

(2) $\Rightarrow$ (3) : Let $B \subseteq A$ and $x \in \overline{B}$.

Then there is a sequence $\{x_n\}$ in B s.t

$$\{x_n\} \longrightarrow x \text{ as } n \to \infty.$$

So, $\{f(x_n)\} \longrightarrow f(x)$ as $n \to \infty$.

Since $\{f(x_n)\}$ is a sequence in $f(B)$, we conclude that $f(x) \in \overline{f(B)}$.

Hence $f(\overline{B}) \subset \overline{f(B)}$.

3 $\Rightarrow$ 4: Let $Z \subset \mathbb{R}$ be a closed set.

Since $f(f^{-1}(Z)) \subset Z$ and Z is closed, we get by applying 3 that

$$f\left(\overline{f^{-1}(Z)}\right) \subset \overline{f(f^{-1}(Z))}$$
$$\subset \overline{Z} = Z.$$

and hence

$$\overline{f^{-1}(Z)} \subset f^{-1}(Z),$$

which implies $f^{-1}(Z)$ is closed.

④ $\Rightarrow$ ⑤: Let $V \subset \mathbb{R}$ be open.

Since $\mathbb{R} \setminus V$ is closed, by ④

we have

$$f^{-1}(\mathbb{R} \setminus V) \text{ is closed.}$$

It follows that

$$A \setminus f^{-1}(V) = f^{-1}(\mathbb{R} \setminus V)$$

is closed

Hence $f^{-1}(V)$ is open in A.

5 $\Rightarrow$ 1 : Let $a \in A$ and $\varepsilon > 0$.

Consider that $V = N_\varepsilon(f(a))$,

which is open in $\mathbb{R}$.

Since $f^{-1}(V)$ is open by 5,

and since $a \in f^{-1}(V)$ there

must exist $\delta > 0$ s.t

$$N_\delta(a) \subset f^{-1}(V).$$

This implies that

$$f(N_\delta(a)) \subset V = N_\varepsilon(f(a)).$$

Since $a \in A$ is arbitrary, f is continuous.

Examples:

① The Constant function

$$f(x) = 1 \quad \text{on } \mathbb{R}$$

is a continuous function.

Let $\{x_n\} \longrightarrow 1$ in $\mathbb{R}$ then

$$\{f(x_n)\} = \{1, 1, 1, 1, \dots\}$$

$$\longrightarrow 1 = f(1).$$

② The function $g(x) = x$ on $\mathbb{R}$

is a continuous function.

(3) Define $h : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$h(x) = x^2, \ \forall x \in \mathbb{R}.$$

Suppose $\{x_n\} \longrightarrow x$ in $\mathbb{R}$

then $\{x_n^2\} \longrightarrow x^2$ as $n \to \infty$.

That is, $\{h(x_n)\} \longrightarrow h(x)$ as $n \to \infty$.

Hence 'h' is a Continuous function.

(4) Define

$$f : \{x \in \mathbb{R} \,/\, x > 0\} \longrightarrow \mathbb{R}$$

by

$$f(x) = \frac{1}{x}, \quad x > 0.$$

Suppose $\{x_n\}$ is a Sequence with $x_n > 0 \ \forall n \in \mathbb{N}$ and $x_n \longrightarrow x \ (x>0)$, then

$$f(x_n) = \frac{1}{x_n} \longrightarrow \frac{1}{x} = f(x)$$

as $n \to \infty$.

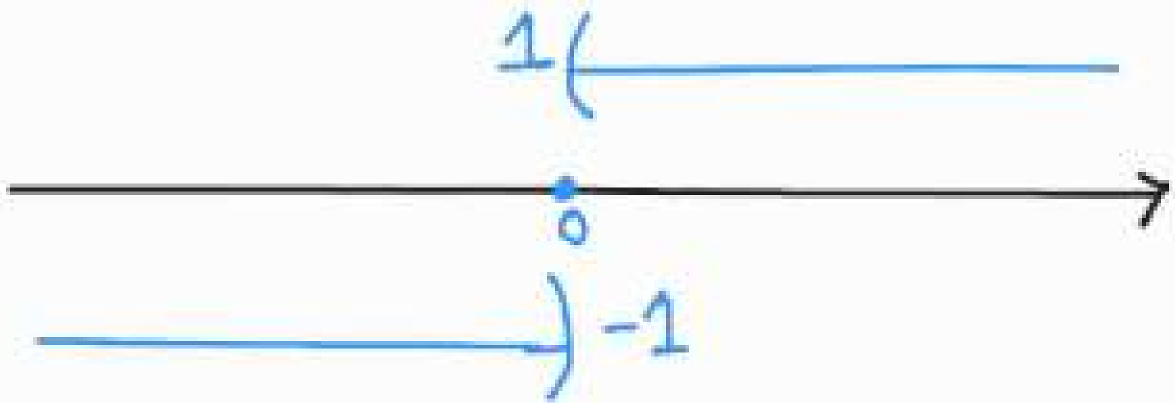
Therefore, f is Continuous.

5 "Signum function" defined by

$$\operatorname{Sgn}(x) = \begin{cases} +1 & \text{for } x>0 \\ 0 & \text{for } x=0 \\ -1 & \text{for } x<0. \end{cases}$$

Note that $\operatorname{sgn}(x) = \dfrac{x}{|x|}$,

for $x \neq 0$.



We show that $\operatorname{sgn}(x)$ is not a continuous function.

Let $x_n = \dfrac{(-1)^n}{n}$ for $n \in \mathbb{N}$

then the sequence $\{x_n\}_{n=1}^{\infty}$

converges to zero. i.e.,

$$\{x_n\} \longrightarrow 0 \quad \text{as } n \to \infty$$

However, $\text{sgn}(x_n) = (-1)^n$,

$n \in \mathbb{N}$.

and so, the sequence

$\{\text{sgn}(x_n)\}_{n=1}^{\infty}$ does NOT converge

Hence $\text{sgn}(x)$ function is

discontinuous at $x=0$.

As a result, signum function

$\text{sign}(x)$ is NOT continuous.

(6) Define $f: \mathbb{R} \longrightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Let $x \in \mathbb{Q}$, then we can choose a sequence $\{x_n\}$ of irrational numbers that converges to x.

Since $f(x_n) = 0 \quad \forall n \in \mathbb{N}$, we have

$$\{f(x_n)\}_{n=1}^{\infty} \longrightarrow 0 \quad \text{but}$$

$$f(0) = 1$$

Therefore, f is NOT continuous

at any rational number x.

Similarly, if $y \in \mathbb{R} \setminus \mathbb{Q}$ then we choose a sequence $\{y_n\}$ of rational numbers that converges to y.

Since $f(y_n) = 1 \quad \forall\, n \in \mathbb{N}$, we have that

$$\{f(y_n)\}_{n=1}^{\infty} \longrightarrow 1 \quad \text{but}$$

$$f(y) = 0$$

Therefore, f is NOT continuous at any irrational number

Since every real number is either rational or irrational, we deduce that f is NOT continuous at any point in $\mathbb{R}$.

The function 'f' defined in example 6 is called Dirichlet's discontinuous function (introduced by P.G.L. Dirchlet) in 1829.

Examples

(7) Let $A = \{x \in \mathbb{R} \mid x > 0\}$

Define $h : A \longrightarrow \mathbb{R}$ by

$h(x) = 0$ for any irrational $x \in A$.

$h\left(\frac{m}{n}\right) = \frac{1}{n}$ for any $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$.

Now we will show that h is continuous at every irrational number in A, and is discontinuous at every rational number in A.

If $x > 0$ is a rational number then we can choose a sequence $\{x_n\}$ of irrational numbers s.t $\{x_n\} \longrightarrow x$ as $n \to \infty$.

From the definition of 'h',

$h(x) > 0$ and $\{h(x_n)\} \longrightarrow 0$

So, h is discontinuous at x.

On the other hand, if x is an irrational number and $\varepsilon > 0$, then $\exists$ $N \in \mathbb{N}$ s.t

$$\frac{1}{N} < \varepsilon.$$

There are only finitely many rationals with denominator less than N in the interval $(x-1, x+1)$.

Hence $\delta > 0$ can be choosen so small that $(x-\delta, x+\delta)$ contains no rational numbers with denominator less than N.

It follows that for $y \in A$ with

$|y-x| < \delta$ we have

$$|h(y)-h(x)| = |h(x)| \leq \frac{1}{N} < \varepsilon.$$

Thus h is continuous at the irrational number x.

The function h defined in (7) is called Thomae's function. This function was introduced in 1875 by K.J. Thomae

Remark:

Sometimes a function $f: A \to \mathbb{R}$ is not continuous at a point 'b' because it is not defined at the point b. However if 'f' has a limit L at the point 'b' then the function

$$F : A \cup \{b\} \longrightarrow \mathbb{R} \text{ by}$$

$$F(x) := \begin{cases} L & ; x = b \\ f(x) & ; x \in A. \end{cases}$$

is continuous at b.

<u>Eg:</u>

① Define $g : \mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R}$ by

$$g(x) = \sin\left(\frac{1}{x}\right),$$

for $x \neq 0$.

The function g is not defined at $x = 0$.

Now we show that 'g' does NOT have limit at $x=0$.

We will construct two sequences $\{x_n\}$ and $\{y_n\}$ with

$$x_n \neq 0 \,, \quad y_n \neq 0 \quad \forall\, n \in \mathbb{N}$$

such that

$$\lim x_n = 0 \,, \quad \lim y_n = 0 \quad \text{but}$$

$$\lim \{g(x_n)\} \neq \lim \{g(y_n)\}.$$

Let $x_n = \dfrac{1}{n\pi}$ for $n \in \mathbb{N}$

and $y_n = \dfrac{1}{2n\pi + \pi/2}$

then $\lim x_n = 0$, $\lim y_n = 0$.

Further,

$$\{g(x_n)\} = \{\sin(n\pi)\}_{n=1}^{\infty}$$

$$\longrightarrow 0$$

and

$$\{g(y_n)\} = \{\sin(\pi/2 + 2n\pi)\}_{n=1}^{\infty}$$

$$= \{1, 1, 1, 1, \cdots\}$$

$$\longrightarrow 1 \quad , \quad \text{as } n \to \infty$$

$$\lim \{g(x_n)\} \neq \lim \{g(y_n)\}$$

$\Rightarrow$ $\lim\limits_{x \to 0} \sin\left(\frac{1}{x}\right)$ does NOT exist.

Thus there is no value that we can assign at $x=0$.

Hence g can not have an extension that is continuous at $x=0$.

Lecture 6 DATE : 27.01.2025

Theorem [106]

Let $A \subseteq \mathbb{R}$ and $f, g : A \to \mathbb{R}$ be continuous at a point "a" $\in A$

then

① $f+g$, $f-g$, fg and $\alpha \cdot f$ are continuous at "a".

(Here $\alpha \in \mathbb{R}$).

② If h is continuous at 'a' and $h(x) \neq 0$ for all $x \in A$ then the quotient $f(x)/h(x)$ is continuous at 'a'.

Proof. (1) Given that both f, g are continuous at 'a'.

Suppose $\{x_n\} \longrightarrow a$ in A

then $\{f(x_n)\} \longrightarrow f(a)$

and $\{g(x_n)\} \longrightarrow g(a)$.

as $n \to \infty$

It follows that

$$(f+g)(x_n) = f(x_n) + g(x_n) \longrightarrow f(a) + g(a)$$

i.e., $$(f+g)(x_n) \longrightarrow (f+g)(a)$$

and so,

$(f+g)$ is continuous at 'a'.

Similarly,

$$(f-g)(x_n) = f(x_n) - g(x_n)$$
$$\longrightarrow f(a) - g(a)$$
$$= (f-g)(a),$$

as $n \to \infty$

$$(f \cdot g)(x_n) = f(x_n) \cdot g(x_n)$$
$$\longrightarrow f(a) \cdot g(a)$$
$$= (f \cdot g)(a)$$

$$(\alpha \cdot f)(x_n) = \alpha \cdot f(x_n)$$

$$\longrightarrow \alpha \cdot f(a)$$

$$= (\alpha \cdot f)(a).$$

This shows that

$f+g$, $f-g$, $f \cdot g$, $\alpha \cdot f$

are continuous at 'a'.

(2) Assume that $h : A \to \mathbb{R}$ is continuous at 'a' and $h(x) \neq 0 \ \forall x \in A$.

If $\{x_n\} \longrightarrow a$ in A

then

$$\left(\frac{f}{h}\right)(x_n) = \frac{f(x_n)}{h(x_n)} \longrightarrow \frac{f(a)}{h(a)} = \left(\frac{f}{h}\right)(a)$$

as $n \to \infty$.

$\therefore$ $\frac{f}{h}$ is continuous at 'a'.

Remark:

① If $h : A \longrightarrow \mathbb{R}$ is continuous on A and $f : A \longrightarrow \mathbb{R}$ is continuous on A, then one can define quotients of functions as below:

First, let $A_1 = \{ x \in A \mid h(x) \neq 0 \}$ then it is clear to see both f and h are continuous on A_1. Now we define quotient $\frac{f}{h}$ on A_1 by

$$\left(\frac{f}{h}\right)(x) := \frac{f(x)}{h(x)}, \quad \forall\, x \in A_1.$$

Then $\left(\frac{f}{h}\right)$ is a continuous function on A_1.

<u>Examples</u>

① Polynomial functions.

$$p(x) = \alpha_k x^k + \alpha_{k-1} x^{k-1} + \cdots + \alpha_1 x + \alpha_0$$

for all $x \in \mathbb{R}$, then we see that $p(x)$ is continuous.

Let $a \in \mathbb{R}$ and $\{x_n\} \to a$ in $\mathbb{R}$. Then

$$p(x_n) = \alpha_k x_n^k + \alpha_{k-1} x_n^{k-1} + \cdots$$
$$\cdots + \alpha_1 x_n + \alpha_0$$
$$\longrightarrow \alpha_k a^k + \alpha_{k-1} a^{k-1} + \cdots$$
$$+ \alpha_1 a + \alpha_0$$
$$= p(a).$$

i.e.,

$$p(x_n) \longrightarrow p(a) \quad \text{as} \quad n \to \infty.$$

p is continuous at every point $a \in \mathbb{R}$

Thus polynomial function $P(x)$ is continuous on $\mathbb{R}$.

② Rational functions.

If p and q are polynomial functions on $\mathbb{R}$, then there are at most a finite number $c_1, c_2 \ldots c_m$ of real roots of q.

If $x \notin \{c_1, c_2, \ldots, c_m\}$ then $q(x) \neq 0$ and define the rational function

$$r(x) := \frac{p(x)}{q(x)},$$

$$x \notin \{c_1, c_2, \ldots c_m\}$$

Clearly, the rational function $r(x)$ is continuous at every point $x \notin \{c_1, c_2, \ldots c_m\}$

(3) $\sin(x)$, $\cos(x)$ are continuous functions on $\mathbb{R}$

First, note that for any real number $\theta \in \mathbb{R}$

$$|\sin(\theta)| \leq |\theta| \quad \text{and}$$

$$|\cos(\theta)| \leq 1.$$

Now, for any $x, y \in \mathbb{R}$ we can write $\sin(x) - \sin(y)$ as,

$\sin(x) - \sin(y)$

$$= \sin\left[\frac{x+y}{2} + \frac{x-y}{2}\right] - \sin\left[\frac{x+y}{2} - \frac{(x-y)}{2}\right]$$

$$= 2\cos\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right).$$

Let $a \in \mathbb{R}$ then

$$|\sin(x) - \sin(a)|$$

$$= \left|2\cos\left(\frac{x+a}{2}\right)\sin\left(\frac{x-a}{2}\right)\right|$$

$$\leq 2\left|\frac{1}{2}(x-a)\right|$$

$$= |x-a|$$

For every $\varepsilon > 0$ choose $\delta > 0$ s.t

$|\sin(x) - \sin(a)| < \varepsilon$ whenever $|x-a| < \delta$.

Thus $\sin(x)$ is continuous at 'a'.
Since $a \in \mathbb{R}$ is arbitrary point,
we conclude that $\sin(x)$ is
continuous on $\mathbb{R}$.

Similarly, one can show that
$\cos(x)$ is continuous on $\mathbb{R}$.
The proof uses the following identity

$$\boxed{\cos(x) - \cos(y) = -2\sin\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)}$$

(4) The functions $\tan(x)$, $\cot(x)$,
$\sec(x)$, $\operatorname{cosec}(x)$ are continuous
where they are defined.

For instance, $\cot(x) = \dfrac{\cos(x)}{\sin(x)}$

It is defined when $\sin(x) \neq 0$

i.e., $x \neq n\pi$, $n \in \mathbb{Z}$.

Since $\cos(x)$, $\sin(x)$ are continuous on $\mathbb{R}$ and the rational function

$$\cot(x) = \frac{\cos(x)}{\sin(x)} \text{ is defined}$$

on $\mathbb{R} \setminus \{n\pi \mid n \in \mathbb{Z}\}$,

we conclude that $\cot(x)$ is a continuous function on

$$\mathbb{R} \setminus \{n\pi \mid n \in \mathbb{Z}\}.$$

EX: Show that if $f : A \to \mathbb{R}$ is continuous on A then

$|f|(x) = |f(x)|$ & $\sqrt{f}(x) = \sqrt{f(x)}$.

$\forall x \in A$, are continuous on A.

Composition of continuous functions

Theorem [107] Let $A, B \subseteq \mathbb{R}$, let $f: A \longrightarrow \mathbb{R}$ and $g: B \longrightarrow \mathbb{R}$ be functions such that $f(A) \subseteq B$.

1. If f is continuous at a point $a \in A$ and g is continuous at $b = f(a) \in B$ then the composition $g \circ f: A \longrightarrow \mathbb{R}$ continuous at "a".

2. If f is continuous on A and g is continuous on B then $g \circ f$ is continuous on A.

Proof. Given that $f: A \longrightarrow \mathbb{R}$ and $g: B \longrightarrow \mathbb{R}$ with $f(A) \subseteq B$.

① Let W be an open subset of $\mathbb{R}$ containing the point $(g \circ f)(a)$

$$\Rightarrow \quad g(f(a)) \in W$$

$$\Rightarrow \quad f(a) \in g^{-1}(W).$$

Since g is continuous at $f(a)$ and W is an open set containing $g(f(a))$ we have that

$$g^{-1}(W) \text{ is open in } B.$$

Now use the fact that f is continuous at a and

$g^{-1}(W)$ is an open set containing $f(a)$ then

$f^{-1}(g^{-1}(W))$ is open in A.

Hence $(g\circ f)^{-1}(W)$ is open in A.

Therefore, $g\circ f$ is continuous at 'a'.

Eg: $g(x) = \sin(x)$ and

$$f(x) = \frac{1}{x}, \quad x \neq 0.$$

$$(g\circ f)(x) = \sin\left(\frac{1}{x}\right), \quad x \neq 0$$

is continuous $(\forall x \neq 0)$

Lecture 7 DATE : 28.01.2025

Definition: A function $f : A \to \mathbb{R}$ is said to be **bounded** on A if there exists a constant $M > 0$ such that $|f(x)| \leq M$, for all $x \in A$.

- In other words, a function 'f' is bounded on a set if its range is a bounded subset of $\mathbb{R}$.

- A function $g : A \to \mathbb{R}$ is **NOT bounded** on A if for any $M > 0$ $\exists$ $x_M \in A$ such that

$$|f(x_M)| > M$$

In this case, g is called "unbounded".

<u>Example:</u>

Define $g : (0,\infty) \longrightarrow \mathbb{R}$ by

$$g(x) = \frac{1}{x}, \ \forall x > 0.$$

Clearly, g is an unbounded function on A because for any $M>0$ we can take a point $x_M = \frac{1}{(M+1)} \in (0,\infty)$ and it satisfy that

$$f(x_M) = M+1 > M.$$

<u>Remark:</u> The above example shows that continuous function <u>need not</u> be bounded.

Ex. Let $I = [a, b]$ be a closed bounded interval. If the $f : I \to \mathbb{R}$ is continuous then f is bounded on I.

- Here the interval I must be bounded
  For instance, $f(x) = x$ on $[0, \infty)$ is continuous but unbounded.
- Here the interval I must be closed.
  For example, $g(x) = \frac{1}{x}$ on $(0, 1]$ is continuous but NOT bounded.
- Here the function must be continuous
  For example, $h : [0, 1] \to \mathbb{R}$ by

$$h(x) = \begin{cases} 1 & \text{if } x = 0 \\ \frac{1}{x} & \text{if } x \in (0, 1] \end{cases}$$

The interval $[0,1]$ is closed and bounded

but h is NOT continuous and unbounded.

The maximum and minimum

<u>Definition</u>: Let $A \subseteq \mathbb{R}$ and $f: A \longrightarrow \mathbb{R}$.

We say that f has an <u>absolute maximum</u> on A if $\exists$ a point $x^* \in A$ s.t

$$f(x^*) \geq f(x) \quad \forall x \in A.$$

We say that f has an <u>absolute minimum</u> if $\exists$ a point $x_* \in A$ s.t

$$f(x_*) \leq f(x) \quad \forall x \in A.$$

Remark: ① A continuous function 'f' on a set A does not necessarily have an absolute maximum or absolute minimum on the set A.

For example, take $A = (0, \infty)$ and

$$f(x) = \frac{1}{x} \quad \forall x \in A.$$

then f is continuous.

Suppose $\exists\ x^* \in (0, \infty)$ s.t

$$f(x^*) \geq f(x) \quad \forall x \in (0, \infty)$$

$$\Rightarrow \quad \frac{1}{x^*} \geq \frac{1}{x} \quad \forall x \in (0, \infty)$$

$$\Rightarrow \quad x \leq x^* \quad \forall x \in (0, \infty)$$

Such an x^* does NOT exist in $(0, \infty)$

Thus f has no absolute maximum.

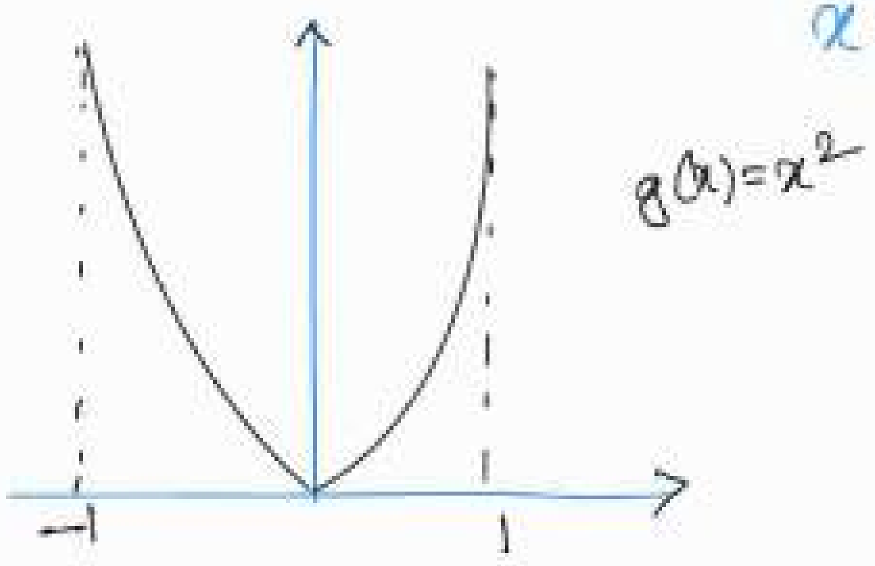
Similarly, f has no absolute minimum on $A = (0, \infty)$.

② If a function has absolute maximum at a point x^*, then this point is not necessarily unique.

For example, consider $A = [-1, 1]$ and $g(x) = x^2$, $\forall x \in A$.

'g' has absolute maximum at $x = \pm 1$.

and has absolute minimum at $x = 0$.



Theorem [108] (Maximum-minimum theorem)

Let $I = [a, b]$ be a closed bounded interval and $f : I \longrightarrow \mathbb{R}$ be continuous on I. Then f has an absolute maximum and absolute minimum on I.

Proof. Consider the set

$$f(I) = \{ f(x) \mid x \in I \}.$$

Since f is continuous, we know that $f(I)$ is bounded in $\mathbb{R}$.

Let $s^* := \sup f(I)$ and

$$s_* := \inf f(I)$$

Claim: $\exists$ $x^*, x_* \in I$ such that

$$s^* = f(x^*) \quad , \quad s_* = f(x_*).$$

since $s^* = \sup f(I)$, for $n \in \mathbb{N}$

$\exists$ $x_n \in I$ s.t

$$s^* - \frac{1}{n} < f(x_n) \leq s^* \quad \text{—} (1)$$

Here $\{x_n\}$ is bounded as I a bounded interval.

By Bolzano-Weierstrass theorem there is a subsequence $\{x_{n_k}\}$ of $\{x_n\}$ that converges to some (say) x^*.

Since I is closed, $x^* \in I$.

From the continuity of f we see that $\{f(x_{n_k})\} \longrightarrow f(x^*)$

as $n_k \longrightarrow \infty$.

It follows from the equation ① that

$$s^* - \frac{1}{n_k} < f(x_{n_k}) \leq s^* \quad \forall\, k \in \mathbb{N}$$

As $n_k \longrightarrow \infty$ we get

$$f(x^*) = s^* = \operatorname{Sup} f(I).$$

$\therefore$ f has an absolute maximum at x^*.

Similar proof works for absolute minimum.

Location of roots

Theorem [109] : Let $I = [a,b]$ and let $f : I \to \mathbb{R}$ be continuous on I. If $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$ then $\exists\ c \in (a,b)$ such that $f(c) = 0$.

Proof. Assume that $f(a) < 0 < f(b)$

We will generate a sequence of intervals by successive bisections.

Let $I_1 := [a_1\ b_1]$ with $a_1 = a$, $b_1 = b$.

and $p_1 = \dfrac{a_1 + b_1}{2}$.

If $f(p_1)=0$ then take $c=p_1$ as desired.

If $f(p_1)\neq 0$ then either

$$f(p_1)>0 \quad \text{or} \quad f(p_1)<0.$$

Suppose $f(p_1)>0$, then we set

$$a_2:=a_1 \, , \quad b_2:=p_1$$

On the other hand when $f(p_1)<0$ we set

$$a_2=p_1 \, , \quad b_2:=b_1$$

In either case, we let $I_2:=[a_2,b_2]$

then we have

$$I_2 \subset I_1 \quad \text{and} \quad f(a_1)<0, \; f(b_2)>0.$$

We continue this bisection process

to obtain the intervals

$I_1, I_2, \ldots, I_k$ with

$f(a_k) < 0$ and $f(b_k) > 0$.

and $p_k := \frac{1}{2}(a_k + b_k)$.

If $f(p_k) = 0$, we take $c := p_k$ and it is done. If $f(p_k) > 0$ we set $a_{k+1} := a_k$, $b_{k+1} := p_k$, while if $f(p_k) < 0$, we set $a_{k+1} := p_k$, $b_{k+1} := b_k$

In either case, we let

$$I_{k+1} := [a_{k+1}, b_{k+1}]$$ then

$$I_{k+1} \subseteq I_k$$ and $f(a_{k+1}) < 0$,

$f(b_{k+1}) > 0$.

If the process terminates by locating a point p_n such that $f(p_n)=0$ then we are done.

If the process does not terminate, then we obtain a nested sequence of closed bounded intervals

$I_n=[a_n,b_n]$ such that

$\forall n\in\mathbb{N}$, $f(a_n)<0$ and $f(b_n)>0$.

and $b_n-a_n = \dfrac{b-a}{2^{n-1}}$.

From the nested intervals property $\exists\ c\in\bigcap_{n=1}^{\infty} I_n$. Here $c\in[a_n,b_n]$ for each $n\in\mathbb{N}$.

i.e., $0\le c-a_n\le b_n-a_n = \dfrac{b-a}{2^{n-1}}$,

and

$$0 \leq b_n - c \leq b_n - a_n = \frac{b-a}{2^{n-1}}$$

Hence, it follows that

$$\lim \{a_n\} = c = \lim \{b_n\}$$

Since f is continuous at 'c',

we get

$$\lim \{f(a_n)\} = f(c) = \lim \{f(b_n)\}$$

Using the fact that $f(a_n) < 0 \ \forall n \in \mathbb{N}$

implies that $\underline{f(c) = \lim \{f(a_n)\} \leq 0}$.

Similarly, using the fact that

$f(b_n) > 0 \ \forall n \in \mathbb{N}$ implies that

$$\underline{f(c) = \lim \{f(b_n)\} \geq 0}.$$

Thus, $\boxed{f(c) = 0}$, which means that 'c' is a root of 'f'.

Theorem [110] (Bolzano's Intermediate Value theorem)

Let I be an interval and let $f: I \longrightarrow \mathbb{R}$ be continuous on I.

If $a, b \in I$ and $f(a) < r < f(b)$ then there exists a point $c \in I$ between 'a' and 'b' such that

$$f(c) = r.$$

Proof: Suppose that $a < b$ and let

$$g(x) := f(x) - r$$

then

$$g(a) < 0 < g(b).$$

Since g is a continuous function on I, by location of roots theorem $\exists\, c \in I$ such that $a < c < b$

and $g(c) = 0$

$$\Rightarrow \quad f(c) = \gamma.$$

If $b < a$, then we take

$$h(x) := \gamma - f(x)$$

so that $h(b) < 0 < h(a)$.

Since 'h' is a continuous on I, by location of roots theorem.

$\exists\, c \in I$ such that

$b < c < a$ and

$h(c) = 0$.

Therefore, $\boxed{f(c) = \gamma.}$

<u>Remark</u>: For a closed and bounded interval I of $\mathbb{R}$ and a continuous function $f : I \longrightarrow \mathbb{R}$, we have that

$$f(I) := \{ f(x) : x \in I \}$$ is a closed and bounded interval.

<u>Proof</u>: If we let $m = \operatorname{Inf} f(I)$ and $M = \operatorname{Sup} f(I)$. From the maximum-minimum theorem, we have

$m, M \in f(I)$.

Moreover, $f(I) \subseteq [m, M]$.

If $m < r < M$ then $\exists$ $c \in I$ s.t $f(c) = r$. That is, $r \in f(I)$ and we conclude that $[m, M] \subseteq f(I)$.

Therefore, $f(I) = [m, M]$.

- If $I = [a,b]$ and $f : I \longrightarrow \mathbb{R}$ we have shown that $f(I) = [m, M]$, this may not be always same as $[f(a), f(b)]$.

For example, see $f(x) = \frac{1}{x^2+1}$ on $[-1, 2]$.

Theorem [111] : Let I be an interval and let $f : I \longrightarrow \mathbb{R}$ be continuous on I. Then the set $f(I)$ is an interval.

Proof. Let $\alpha, \beta \in f(I)$ with $\alpha < \beta$.

then there exist points $a, b \in I$ s.t $\alpha = f(a)$, $\beta = f(b)$.

If $\alpha < \gamma < \beta$ then by the intermediate value theorem $\exists\ c \in I$ s.t $f(c) = \gamma$

This follows that

$$[\alpha, \beta] \subseteq f(I).$$

Hence $f(I)$ is an interval.

Uniform continuity

Recall that a function $f: A \to \mathbb{R}$ is continuous at a point $a \in A$

if and only if

for a given $\varepsilon > 0$ $\exists\ \delta > 0$ such that

$$|f(x) - f(a)| < \varepsilon \quad \text{whenever} \quad |x - a| < \delta.$$

Here 'δ' depends on both 'ε' and 'a'.

The fact that δ depends on 'a' is a reflection of the fact that the function f may change its values rapidly near certain points and slowly near other points

- In some cases, the number $\delta > 0$ can be choosen to be independent of the point $a \in A$ and to depend only on ε.

For example, if $f(x) = 2x \quad \forall x \in \mathbb{R}$

$$|f(x) - f(a)| = 2|x-a|$$

So, for $\varepsilon > 0$ and for any $a \in A$

We can choose $\delta = \varepsilon/2$

Eg ②: Let $A = \{x \in \mathbb{R} \mid x > 0\}$

and $g : A \longrightarrow \mathbb{R}$ by

$$g(x) = \frac{1}{x}, \quad \forall x \in A$$

For $a \in A$,

$$g(x) - g(a)$$

$$= \frac{1}{x} - \frac{1}{a}$$

$$= \frac{a - x}{ax}.$$

Let $\varepsilon > 0$ and if we take

$$\delta := \inf\left\{\frac{1}{2}a, \frac{1}{2}a^2\varepsilon\right\}$$

Suppose

$$|x-a| < \delta$$

when $\quad |x-a| < \frac{1}{2}a$

$$\Rightarrow \quad \frac{1}{2}a < x < \frac{3}{2}a$$

$$\Rightarrow \quad \frac{1}{x} < \frac{2}{a}.$$

Thus if $|x-a| < \frac{1}{2}a$ then

$$|g(x)-g(a)| = \frac{|a-x|}{|ax|}$$

$$< \frac{2}{a^2}|x-a|.$$

As a result, if $|x-a| < \delta$

then

$$|g(x)-g(a)| < \frac{2}{a^2}\frac{1}{2}a^2\varepsilon = \varepsilon$$

Here $\delta = \min\left\{\frac{a}{2}, \frac{a^2\varepsilon}{2}\right\}$ depends on both 'a' and ε. So we denote it by
$$\delta(\varepsilon, a) = \min\left\{\frac{a}{2}, \frac{a^2\varepsilon}{2}\right\}$$

If we wish to consider for all $a \in A$, then $\delta(\varepsilon, a)$ does not lead to one value $\delta(\varepsilon) > 0$ because
$$\inf\{\delta(\varepsilon, a) \mid a > 0\} = 0.$$

In fact there is no way of choosing one value of δ depends only on $\varepsilon > 0$ for the function $g(x) = \frac{1}{x}$.

MTH 205: ANALYSIS IN ONE VARIABLE

Lecture 9 DATE : 11.02.2025

Definition : Let $A \subseteq \mathbb{R}$ and let $f : A \longrightarrow \mathbb{R}$ be a function. We say that f is uniformly continuous on A if for every $\varepsilon > 0$ there is a $\delta > 0$ (depends only on ε) such that

$$|f(x) - f(a)| < \varepsilon$$

whenever $|x - a| < \delta$.

- If $f : A \longrightarrow \mathbb{R}$ is uniformly continuous on A then f is continuous on A. However, the converse does not hold true.

For eg: $g(x) = \frac{1}{x}$, $x>0$

'g' is continuous on $A = \{x \mid x>0\}$

but NOT uniformly continuous.

Criteria (For non-uniform continuity)

Let $A \subseteq \mathbb{R}$ and let $f: A \rightarrow \mathbb{R}$ then the following statements are equivalent:

① f is NOT uniformly continuous on A.

② There exists an $\varepsilon > 0$ such that for every $\delta > 0$ there are points x_δ, a_δ in A such that

$$|x_\delta - a_\delta| < \delta \quad \text{and}$$

$$|f(x_\delta) - f(a_\delta)| \geq \varepsilon.$$

③ There exists an $\varepsilon > 0$ and two sequences $\{x_n\}$ and $\{a_n\}$ in A such that

$$\lim_{n\to\infty} (x_n - a_n) = 0 \quad \text{and}$$

$$|f(x_n) - f(a_n)| \geq \varepsilon, \ \forall n \in \mathbb{N}.$$

For instance, $g(x) = \frac{1}{x}$, $x > 0$

If $x_n = \frac{1}{n}$ and $a_n = \frac{1}{n+1}$

then $\lim\limits_{n\to\infty} (x_n - a_n)$

$$= \lim_{n\to\infty} \frac{1}{n(n+1)}$$

$$= 0$$

but $|g(x_n) - g(a_n)| = 1 \ \forall n \in \mathbb{N}.$

Theorem [112]

Let I be a closed bounded interval and let $f: I \longrightarrow \mathbb{R}$ be continuous on I. Then f is uniformly continuous on I.

Proof. Assume that f is NOT uniformly continuous, then there exists $\varepsilon > 0$ and two sequences $\{x_n\}$ and $\{a_n\}$ in I such that $\lim_{n \to \infty} |x_n - a_n| = 0$ and $|f(x_n) - f(a_n)| > \varepsilon$, $\forall n \in \mathbb{N}$.

Since I is bounded, $\{x_n\}$ is bounded, by Bolzano-Weierstrass theorem $\exists \{x_{n_k}\}$ of $\{x_n\}$ s.t $\{x_{n_k}\} \longrightarrow z$ (say)

Since I is closed, $\delta \in I$.

Now consider $\{a_{n_k}\}$ corresponding to $\{x_{n_k}\}$, we have

$$|a_{n_k} - \delta| \le |a_{n_k} - x_{n_k}| + |x_{n_k} - \delta| \longrightarrow 0$$

Since f is continuous,

$$\{f(x_{n_k})\} \longrightarrow f(\delta) \quad \&$$

$$\{f(a_{n_k})\} \longrightarrow f(\delta)$$

but this is NOT possible since $|f(x_n) - f(a_n)| \geq \varepsilon$. $\forall n \in \mathbb{N}$.

Hence the result.

Definition:

Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$.
If there exists a constant $M > 0$
such that

$$|f(x) - f(a)| \leq M|x - a|$$

for all $x, a \in A$ then f is said to be a <u>Lipschitz function</u>

- In other words,

$$\left| \frac{f(x) - f(a)}{x - a} \right| \leq M$$

for all $x, a \in I$, $x \neq a$.

This means that f is Lipschitz function if and only if the slopes of all the line segments joining two points on the graph of $y=f(x)$ over I are bounded by some $M>0$.

- If $f: A \to \mathbb{R}$ is a Lipschitz function then f is uniformly continuous on A.

$\exists\, M>0$ such that

$$|f(x)-f(a)| \leq M\,|x-a|, \ \forall_{x,a\in A}.$$

For $\varepsilon>0$ choose $\delta = \frac{\varepsilon}{M}$ (depends only on ε)

if $|x-a| < \delta = \frac{\varepsilon}{M}$ then

$$\begin{aligned}|f(x)-f(a)| &\leq M\,|x-a| \\ &< M\cdot\frac{\varepsilon}{M} \\ &= \varepsilon\end{aligned}$$

Therefore, f is uniformly continuous on A.

Remark: A uniformly continuous function not necessarily Lipschitz function.

For instance,

Eg ① $g : [0,2] \longrightarrow \mathbb{R}$

$$g(x) = \sqrt{x} \ , \quad 0 \le x \le 2.$$

Here the domain of 'g' is a closed bounded interval and g is continuous by theorem [112], we see that 'g' is uniformly continuous on $[0,2]$.

Now we show that g is NOT Lipschitz function. Assume $\exists M > 0$ such that

$$|f(x) - f(a)| \le M |x-a|$$

$$\forall x, a \in [0,2]$$

For each $n \in \mathbb{N}$, $a=0$

$$\left| f\left(\frac{1}{n}\right) - f(0) \right|$$

$$\leq M \left|\frac{1}{n}\right|$$

$$\Rightarrow \quad \frac{1}{\sqrt{n}} \leq \frac{M}{n}$$

$$\Rightarrow \quad \sqrt{n} \leq M \quad \text{for all } n \in \mathbb{N}$$

This is a contradiction and so. our assumption is wrong. Thus $g(x) = \sqrt{x}$ on $[0,2]$ is NOT a Lipschitz function.

Eg ②: Now define

$$g : [1, \infty) \longrightarrow \mathbb{R} \text{ by}$$

$$g(x) = \sqrt{x}, \quad 1 \leq x < \infty.$$

For $x, a \in [1, \infty)$, we have

$$|g(x) - g(a)| = |\sqrt{x} - \sqrt{a}|$$

$$= \frac{|x-a|}{\sqrt{x}+\sqrt{a}}$$

$$\leq \frac{1}{2}|x-a|$$

Thus g is a Lipschitz function and so, g is uniformly continuous on $[1, \infty)$.

Is $g(x) = \sqrt{x}$, $x \geq 0$ uniformly continuous?

- Note that $f(x) = \frac{1}{x}$ on $(0,1)$ is not uniformly continuous (but continuous)

However, a continuous function defined on closed bounded interval is uniformly continuous.

<u>Question:</u> Under what conditions is a function uniformly continuous on a bounded open interval?

We will see that a function defined on (a,b) is uniformly continuous if and only if it can be defined at the end points to produce a function that is continuous on the closed interval

Theorem [113] If $f: A \to \mathbb{R}$ is uniformly continuous then f maps every Cauchy sequence in A to a Cauchy sequence in $\mathbb{R}$.

Proof. Since f is uniformly continuous.

for $\varepsilon > 0$ $\exists\, \delta > 0$ s.t

$$|f(x) - f(a)| < \varepsilon \quad \text{for any}$$

$x, a \in A$ with $|x - a| < \delta$.

Let $\{x_n\}$ be a Cauchy sequence in A. For this $\delta > 0$ $\exists\, N_\delta \in \mathbb{N}$ s.t

$$|x_n - x_m| < \delta \quad \text{for } n, m \geq N_\delta.$$

$$\Rightarrow \quad |f(x_n) - f(x_m)| < \varepsilon \quad \text{for } n, m \geq N_\delta$$

This shows that $\{f(x_n)\}$ is a Cauchy Sequence in $\mathbb{R}$.

Theorem [114]

A function $f : (a,b) \longrightarrow \mathbb{R}$ is uniformly continuous if and only if it can be defined at the endpoints a, b such that extended function is continuous on $[a,b]$.

Proof. Suppose f is uniformly continuous on (a,b). Now we will see how to extend f to points a & b.

Let $\{x_n\}$ be a sequence in (a,b) then $\{x_n\}$ is a Cauchy sequence.

It follows that $\{f(x_n)\}$ is a Cauchy sequence in $\mathbb{R}$. Every Cauchy sequence is convergent in $\mathbb{R}$ and hence $\{f(x_n)\}$ converges to some L.

If $\{y_n\}$ is another sequence in (a,b) that converges to a then

$\lim_{n\to\infty}(x_n - y_n) = 0$, so by uniform continuity of 'f' we have

$$\lim_{n\to\infty} f(y_n) = \lim_{n\to\infty} f(y_n) - f(x_n)$$
$$+ \lim_{n\to\infty} f(x_n)$$
$$= 0 + L$$
$$= L.$$

If we define $f(a) = L$ then

f is continuous at 'a'.

By similar arguments, f can be defined at 'b' and so it is continuous at b.

Thus the redefined function 'f' (including the end points) is continuous on $[a,b]$.

Eg: $$f(x) = \sin\left(\frac{1}{x}\right)$$

is NOT uniformly continuous on $(0,b]$ for $b>0$ because the $\lim_{x \to 0} f(x) = \lim_{x \to 0} \sin\left(\frac{1}{x}\right)$ does not exist.

On the other hand,

the function $g(x) = x\sin\left(\frac{1}{x}\right)$

is uniformly continuous on $(0, b]$

for any $b > 0$.

This is because,

$$\lim_{x \to 0} x \sin\left(\frac{1}{x}\right) = 0$$

So, $g(x)$ can be redefined as

$g(0) = 0$ and

$g(x) = x\sin\left(\frac{1}{x}\right)$, $0 < x \leq b$

is continuous ▨

Lecture 10 DATE : 18.02.2025

Remark: Let I be a closed bounded interval and let $f: I \to \mathbb{R}$ be continuous on I. If $\varepsilon > 0$, then there is a step function $s_\varepsilon : I \to \mathbb{R}$ such that

$$|f(x) - s_\varepsilon(x)| < \varepsilon \quad , \forall x \in I.$$

Let $\varepsilon > 0$. Since f is uniformly continuous $\exists$ $\delta > 0$ (depends on $\varepsilon > 0$) s.t

$$|f(x) - f(y)| < \varepsilon \quad \text{whenever}$$

$$|x - y| < \delta.$$

Suppose $I = [a,b]$, then we can choose the largest $m \in \mathbb{N}$ with

$$h := \frac{b-a}{m} < \delta.$$

Now divide the interval $I = [a,b]$ into 'm' disjoint intervals of length 'h' as,

$$I_1 = [a, a+h]$$

$$I_2 = [a+h, a+2h]$$

$$\vdots$$

$$I_k = [a+(k-1)h, a+kh].$$

$$k = 2, 3, 4, \cdots m$$

and $\quad l(I_k) = h < \delta.$

Now define $s_\varepsilon : I \longrightarrow \mathbb{R}$ by

$$s_\varepsilon(x) = f(a+kh) \quad \text{if } x \in I_k.$$

$$k = 2, 3, \cdots m.$$

Here s_ε is constant on each interval and so, s_ε is a step function.

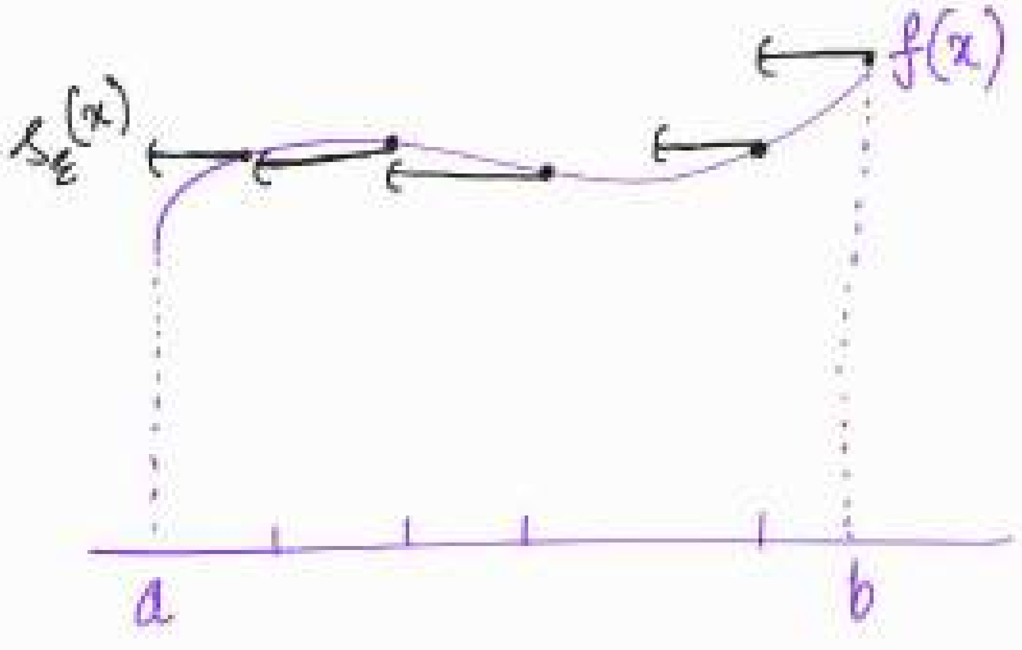
Furthermore, if $x \in I_k$ then

$$\begin{aligned} |f(x) - s_\varepsilon(x)| &= |f(x) - f(a+kh)| \\ &< \varepsilon \end{aligned}$$

because $|x - a - kh| \leq h < \delta$.

Therefore, $|f(x) - s_\varepsilon(x)| < \varepsilon$

for all $x \in I$.



Theorem [115] Let I be a closed bounded interval and let $f: I \to \mathbb{R}$ be continuous on I. If $\varepsilon > 0$ then there is a piecewise linear function $g_\varepsilon : I \to \mathbb{R}$ s.t

$$|f(x) - g_\varepsilon(x)| < \varepsilon \quad \text{for all } x \in I.$$

Proof. Let $\varepsilon > 0$. Since f is uniformly continuous $\exists\ \delta > 0$ (dependes on $\varepsilon > 0$)

if $x, y \in I$ with $|x-y| < \delta$ then

$$|f(x) - f(y)| < \varepsilon.$$

We can choose the largest $m \in \mathbb{N}$ with

$$h := \frac{b-a}{m} < \delta$$

Now divide I into m disjoint intervals of length h as,

$$I_1 = [a, a+h] \quad , \quad I_2 = (a+h, a+2h]$$

$$I_k = (a+(k-1)h \, , \, a+kh],$$

$$k = 2, 3, 4, \cdots m.$$

Define $g_\varepsilon : I \longrightarrow \mathbb{R}$ as

the linear function joining the points $\big(a+(k-1)h, f(a+(k-1)h)\big)$ and $\big(a+kh, f(a+kh)\big)$.

on the interval I_k.

That is, when $x \in I_k$

$$g_\varepsilon(x)$$

$$= f(a+(k-1)h) + \frac{f(a+kh) - f(a+(k-1)h)}{h}$$

$$(x - a - (k-1)h)$$

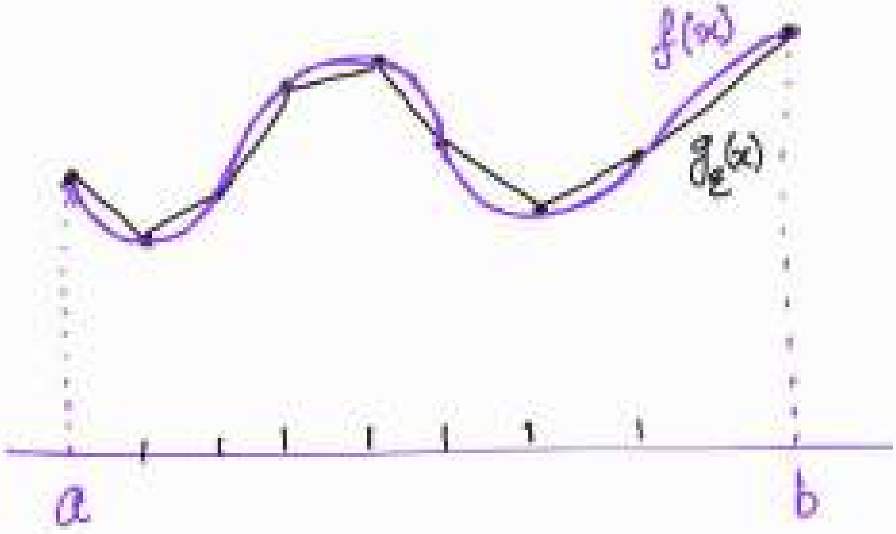
Then g_ε is a continuous piecewise linear function on I and if $x \in I_k$

then

$$|f(x) - g_\varepsilon(x)| < \varepsilon \qquad k = 2, 3 \cdots m$$

Therefore, we conclude that

$$|f(x) - g_\varepsilon(x)| < \varepsilon \quad , x \in I.$$



Theorem [116] Let $I=[a,b]$ and let $f: I \to \mathbb{R}$ be a continuous function. If $\varepsilon > 0$ is given then there exists a polynomial function P_ε s.t

$$|f(x) - P_\varepsilon(x)| < \varepsilon, \quad \forall x \in I.$$

- There are several proofs available for this result. We present one of the most elementary proof below.

Bernstein polynomials

Suppose $f : [0,1] \longrightarrow \mathbb{R}$ is a continuous function. For each $n \in \mathbb{N}$, define

$$B_n(f) = \sum_{k=0}^{n} \binom{n}{k} x^k (1-x)^{n-k} f\left(\frac{k}{n}\right).$$

Here $B_n(f)$ is called the n^{th} Bernstein polynomial for "f".

If we denote the function

$$x \longmapsto x^n \quad \text{on } [0,1]$$

by "x^n" for $n \geq 0$.

then we have the following expressions:

$$B_n(1) = 1,$$

$$B_n(x) = x,$$

$$B_n(x^2) = \frac{(n-1)}{n}x^2 + \frac{1}{n}x$$

$$B_n(x^3) = \frac{(n-1)(n-2)}{n^2}x^3 + \frac{3(n-1)}{n^2}x^2 + \frac{1}{n^2}x$$

$$B_n(x^4) = \frac{(n-1)(n-2)(n-3)}{n^3}x^4 + \frac{6(n-1)(n-2)}{n^3}x^3 + \frac{7(n-1)}{n^3}x^2 + \frac{1}{n^3}x.$$

$$\vdots$$

This implies that

$$\sum_{k=0}^{n} \binom{n}{k} \left(\frac{k}{n} - x\right)^2 x^k (1-x)^{n-k}$$
$$= x(1-x)\frac{1}{n}. \quad \text{— (I)}$$

$$\sum_{k=0}^{n} \binom{n}{k} \left(\frac{k}{n} - x\right)^4 x^k (1-x)^{n-k}$$
$$= x(1-x)\frac{(3n-6)x(1-x)+1}{n^3}.$$

⋮

Theorem [116] (Bernstein Approximation theorem)

Let $f : [0,1] \longrightarrow \mathbb{R}$ be a continuous.

For each $\varepsilon > 0$ $\exists$ $N_\varepsilon \in \mathbb{N}$ such that

$$|f(x) - B_n(f)| < \varepsilon,$$

when $n \geq N_\varepsilon$, for all $x \in [0,1]$.

<u>Proof.</u> Given that f is a continuous function on the closed interval $[a,b]$

So, f is bounded. say

$$|f(x)| \leq M, \quad \forall x \in [0,1].$$

Moreover, f is uniformly continuous

For $\varepsilon > 0$ $\exists$ $\delta > 0$ such that

if $|x-y| < \delta$ then

$$|f(x) - f(y)| < \varepsilon/2 .$$

$$|B_n(f) - f(x)|$$

$$\leq \sum_{k=0}^{n} \binom{n}{k} x^k (1-x)^{n-k} \, |f(k/n) - f(x)|$$

Let $S \subseteq \{0, 1, 2, \cdots n\}$ be such that for every $k \in S$, $|k/n - x| < \delta$

$$\sum_{k \in S} \binom{n}{k} x^k (1-x)^{n-k} |f(k/n) - f(x)|$$

$$< \frac{\varepsilon}{2} \sum_{k \in S} \binom{n}{k} x^k (1-x)^{n-k}$$

$$= \varepsilon/2$$

On the other hand, if $k \in S^c$

then $\quad \delta^2 \leq \left(\frac{k}{n} - x\right)^2$

and so,

$$\delta^2 \sum_{k \in S^c} \binom{n}{k} x^k (1-x)^{n-k} \left| f(k/n) - f(x) \right|$$

$$\leq \sum_{k \in S^c}^{n} \binom{n}{k} \left(\frac{k}{n} - x\right)^2 x^k (1-x)^{n-k} \left| f(k/n) - f(x) \right|$$

$$\leq 2M \frac{x(1-x)}{n}.$$

It follows that

$$\sum_{k \in S^c} \binom{n}{k} x^k (1-x)^{n-k} \left| f(k/n) - f(x) \right| \leq \frac{2M}{n\delta^2}$$

Now choose $N_\varepsilon \in \mathbb{N}$ s.t

$$\frac{2M}{N_\varepsilon \delta^2} < \varepsilon/2$$

$$\Rightarrow \qquad N_\varepsilon > \frac{4M}{\varepsilon \delta^2}.$$

For any $n \geq N_\varepsilon$, we have

$$|B_{N_\varepsilon}(f) - f(x)|$$

$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$$

$$= \varepsilon$$

Therefore, for $\varepsilon > 0$ $\exists N_\varepsilon \in \mathbb{N}$ such that

$$|B_n(f) - f(x)| < \varepsilon \quad , n \geq N_\varepsilon \quad \forall x \in [0,1]$$

<u>Remark</u>: If $f : [a,b] \longrightarrow \mathbb{R}$ is Continuous then define $F : [0,1] \longrightarrow \mathbb{R}$ by

$$F(t) = f\left(a + (b-a)t\right) \quad \forall \; t \in [0,1]$$

Here F is a Continuous function on $[0,1]$, so it can be approximated by Bernstein polynomials as shown in Theorem [116].

As a result, 'f' is approximated by polynomials as mentioned in the Weierstrass approximation theorem. ▨

Sequence of functions

Definition: Let $S \subseteq \mathbb{R}$ and $f_n : S \longrightarrow \mathbb{R}$ be a function $(\forall n \in \mathbb{N})$.

We say that

① $\{f_n\}$ Converges pointwise to a function $f : S \longrightarrow \mathbb{R}$ if

$$\{f_n(x)\} \longrightarrow f(x) \text{ in } \mathbb{R}$$

for each $x \in S$.

② $\{f_n\}$ Converges uniformly to a function $f : S \longrightarrow \mathbb{R}$ if for every $\varepsilon > 0$ $\exists$ $N_\varepsilon \in \mathbb{N}$ s.t

$$|f(x) - f_n(x)| < \varepsilon \quad \text{for } n \geq N_\varepsilon \text{ and } x \in S.$$

③ $\{f_n\}$ is uniformly Cauchy if

for every $\varepsilon > 0$ $\exists$ $N_\varepsilon \in \mathbb{N}$ s.t

$$|f_n(x) - f_m(x)| < \varepsilon \quad \text{for } n, m \geq N_\varepsilon \text{ and } x \in S.$$

<u>Exercise</u>: Let $\{f_n\}$ be a sequence of real-valued functions on 'S'. Show that

① $\{f_n\}$ does NOT converge uniformly to $f: S \longrightarrow \mathbb{R}$

$\Leftrightarrow$ $\exists$ $\varepsilon > 0$ and a strictly increasing sequence $\{n_k\}$ in $\mathbb{N}$ and a sequence $\{x_k\}$ in S s.t

$$|f(x_k) - f_{n_k}(x_k)| \geq \varepsilon$$

for every $k \in \mathbb{N}$.

② $\{f_n\}$ fails to be uniformly Cauchy $\Leftrightarrow$ $\exists\, \varepsilon > 0$, two strictly increasing sequences $\{m_k\}$ and $\{n_k\}$ in $\mathbb{N}$ and a sequence $\{x_k\}$ in S s.t

$$|f(x_{m_k}) - f(x_{n_k})| \geq \varepsilon, \forall k \in \mathbb{N}$$

③ If $\{f_n\}$ converges uniformly to some $f: S \longrightarrow \mathbb{R}$ then $\{f_n\} \longrightarrow f$ pointwise and $\{f_n\}$ is uniformly Cauchy.

Eg: ① Define $f_n: \mathbb{R} \longrightarrow \mathbb{R}$ by

$$f_n(x) = \frac{x}{n} \quad ; \quad x \in \mathbb{R}$$

Here, for each $x \in \mathbb{R}$, the sequence $\{f_n(x)\} = \left\{\frac{x}{n}\right\}$

$$\longrightarrow 0 \quad \text{as } n \to \infty.$$

By defining, $f \equiv 0$ on $\mathbb{R}$

we conclude that $f_n \longrightarrow f$

pointwise.

However, $\{f_n\}$ does NOT

converge uniformly to f.

(why?)

② Let $f_n : [0,1] \longrightarrow \mathbb{R}$ be

defined by

$$f_n(x) = x^n \; ; \quad x \in [0,1]$$

$n \in \mathbb{N}$.

Consider $f : [0,1] \longrightarrow \mathbb{R}$

given by

$$f(x) = \begin{cases} 1 & , \text{ if } x = 1 \\ 0 & \text{ if } 0 \leq x < 1. \end{cases}$$

if $x = 1$, then

$$\{f_n(1)\} = \{1, 1, 1, 1, \ldots\}$$

$$\longrightarrow 1 = f(1) \quad ;$$

if $0 \leq x < 1$, then

$$\{f_n(x)\} = \{x^n\} \longrightarrow 0 = f(x)$$

i.e., $\{f_n\} \longrightarrow f$ point wise.

Now, let $x_n = \left(\frac{1}{2}\right)^{1/n}$, $n \in \mathbb{N}$

then

$$|f(x_n) - f_n(x_n)|$$

$$= \left|0 - \frac{1}{2}\right|$$

$$= \frac{1}{2}, \quad \forall n \in \mathbb{N}$$

By Exercise 1, we conclude that $\{f_n\}$ does NOT converge to f uniformly.

③ Let $f_n : [0,1] \longrightarrow \mathbb{R}$ be defined by

$$f_n(x) = \frac{x^n}{1+x^n}, \ \forall n \in \mathbb{N}$$

Define

$$f(x) = \begin{cases} \frac{1}{2} & \text{if } x=1 \\ 0 & \text{if } 0 \le x < 1 \end{cases}$$

then

$$f_n(x) \longrightarrow f(x)$$

$$\forall \ x \in [0,1]$$

i.e.,

$$\{f_n\} \longrightarrow f \ \text{ pointwise.}$$

If $x_n = \left(\frac{1}{2}\right)^{\frac{1}{n}}$ $\forall n \in \mathbb{N}$

then

$$|f(x_n) - f_n(x_n)|$$

$$= \frac{1}{3} \text{ for every } n \in \mathbb{N}.$$

By Exercise 1, we see that $\{f_n\}$ does NOT converge uniformly

4 Let $f_n : (0,\infty) \longrightarrow \mathbb{R}$ by

$$f_n(x) = \log\left(x + \frac{1}{n}\right)$$

$\forall n \in \mathbb{N}$

Define $f : (0,\infty) \longrightarrow \mathbb{R}$ by

$$f(x) = \log(x)$$

then $\{f_n\} \longrightarrow f$ pointwise.

Is $\{f_n\} \longrightarrow f$ uniformly?

<u>Theorem [117]</u> Let $f_n : S \longrightarrow \mathbb{R}$ be a continuous $(\forall\, n \in \mathbb{N})$, $S \subseteq \mathbb{R}$

① If $\{f_n\}$ Converges uniformly to $f : S \longrightarrow \mathbb{R}$ then

$$f(x) = \lim_{n\to\infty} f_n(x_n) \text{ whenever } \{x_n\} \longrightarrow x \text{ in } S.$$

(2) If $\{f_n\}$ is uniformly Cauchy on a dense subset D of S then $\{f_n\}$ is uniformly Cauchy on S.

Proof. (1) First, note that

$\{f_n\} \longrightarrow f$ uniformly.

It follows that 'f' is continuous (why?)

Let $\{x_n\} \longrightarrow x$ in 'S' and $\varepsilon > 0$.

Then $\exists\ N_\varepsilon \in \mathbb{N}$ s.t

$$|f(x_n) - f(x)| < \varepsilon/2, \ \forall\, n \geq N_\varepsilon$$

and

$$|f(y) - f_n(y)| < \varepsilon/2,$$
$$\forall\, n \geq N_\varepsilon$$

As a result,

$$|f(x) - f_n(x_n)|$$
$$\leq |f(x) - f(x_n)|$$
$$+ |f(x_n) - f_n(x_n)|$$
$$< \varepsilon/2 + \varepsilon/2$$
$$= \varepsilon, \quad \forall\, n \geq N_\varepsilon$$

$$\Rightarrow \quad \lim_{n\to\infty} f_n(x_n) = f(x).$$

(2) Let $x \in S$ then

there is $\{x_n\}$ in D s.t

$x_n \longrightarrow x$ as $n \to \infty$.

For $\varepsilon > 0$ $\exists$ $N_\varepsilon \in \mathbb{N}$ such that

$$| f_n(x_k) - f_m(x_k) | < \varepsilon/3$$

for every $n, m \geq N_\varepsilon$

and for every x_k.

and

$$| f_n(x_k) - f_n(x) | < \varepsilon/3,$$

$$|f_m(x_k) - f_m(x)| < \varepsilon/3$$

for $n, m \geq N_\varepsilon$

$$
\begin{aligned}
|f_n(x) - f_m(x)| &\leq |f_n(x) - f_n(x_k)| \\
&\quad + |f_n(x_k) - f_m(x_k)| \\
&\quad + |f_m(x_k) - f_m(x)| \\
&< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 \\
&= \varepsilon.
\end{aligned}
$$

Definition: Let $S \subseteq \mathbb{R}$ and $\mathcal{F}$ be a family of functions from S to $\mathbb{R}$. We say that

(i) $\mathcal{F}$ is pointwise bounded if $\forall x \in S$, there is $M_x > 0$ s.t

$$|f(x)| \leq M_x, \ \forall f \in \mathcal{F}$$

(ii) $\mathcal{F}$ is uniformly bounded if there is $M > 0$ s.t

$$|f(x)| \leq M, \ \forall f \in \mathcal{F}, \ x \in S.$$

Eg: (1) If $f_n : \mathbb{R} \longrightarrow \mathbb{R}$ is given by $f_n(x) = \frac{x}{n}$

then $\mathcal{F} = \{ f_n : n \in \mathbb{N} \}$

is pointwise bounded

because $(M_x := |x|)$

$$|f_n(x)| = \frac{|x|}{n} \leq |x|,$$

$$\forall f_n \in \mathcal{F}.$$

However, $\mathcal{F}$ is NOT uniformly bounded.

Definition: Let $S \subseteq \mathbb{R}$ and a family $\mathcal{F}$ of functions from S to $\mathbb{R}$ is said to be "equicontinuous" if for every $\varepsilon > 0$ $\exists\, \delta > 0$ s.t

$$f\big((x-\delta, x+\delta)\big) \subseteq \big(f(x)-\varepsilon, f(x)+\varepsilon\big),$$

for every $f \in \mathcal{F}$, $x \in S$.

Remark: Let $\mathcal{F}$ be a family of functions from S to $\mathbb{R}$. Then $\mathcal{F}$ fails to be equicontinuous if and only if $\exists\ \varepsilon > 0$ and $\{x_n\}, \{y_n\}$ in S, a sequence $\{f_n\}$ in $\mathcal{F}$ such that

$$|x_n - y_n| \to 0 \quad \text{and}$$

$$|f_n(x_n) - f_n(y_n)| \geq \varepsilon, \quad \forall n \in \mathbb{N}$$

Note: If $\mathcal{F}$ is equicontinuous

then f is uniformly continuous, $\forall f \in \mathcal{F}$

but the converse need not hold true.

For example, $\mathcal{F} = \{ f_n : n \in \mathbb{N} \}$

where $f_n : [0,1] \longrightarrow \mathbb{R}$

is given by

$$f_n(x) = x^n, \quad x \in [0,1]$$
$$n \in \mathbb{N}$$

Each 'f_n' is a uniformly continuous

However, take $\varepsilon = \frac{1}{2}$

$x_n = 1$, $y_n = \left(\frac{1}{2}\right)^{1/n}$ $n \in \mathbb{N}$

we have

$$|x_n - y_n| = \left|1 - \left(\frac{1}{2}\right)^{1/n}\right|$$

$$\longrightarrow 0$$

as $n \to \infty$.

and

$$|f_n(x_n) - f_n(y_n)|$$

$$= \left|1 - \frac{1}{2}\right| = \frac{1}{2},$$

$\forall\, n \in \mathbb{N}$

Thus, $\mathcal{F} = \{ f_n : n \in \mathbb{N} \}$

is NOT equicontinuous

Lecture 12 DATE : 24.02.2025

Recall that, a family $\mathcal{F}$ of functions from S to $\mathbb{R}$ is said to be "equicontinuous" if

for every $\varepsilon > 0$ $\exists$ $\delta > 0$ s.t

$$f\big((x-\delta, x+\delta)\big) \subset \big(f(x)-\varepsilon, f(x)+\varepsilon\big)$$

for every $f \in \mathcal{F}$, $x \in S$.

Eg: $\mathcal{F} = \{ f_n : n \in \mathbb{N} \}$ where

$f_n : [0,1] \longrightarrow \mathbb{R}$ given by

$$f_n(x) = \frac{x}{n}$$

For $\varepsilon>0$ choose $\delta=\varepsilon>0$

and if, for any $x\in[0,1]$,

$y\in(x-\delta, x+\delta)$

then

$$|f_n(x)-f_n(y)|$$

$$= \left|\frac{x-y}{n}\right|$$

$$< \frac{\varepsilon}{n} < \varepsilon \quad \text{for each } n\in\mathbb{N}$$

This shows that $\mathcal{F}=\{f_n : n\in\mathbb{N}\}$

is an equicontinuous family.

Remark: (1) $\mathcal{F}$ fails to be an equicontinuous family

$\Leftrightarrow$ $\exists\ \varepsilon>0$, a sequence

$\{x_n\}, \{y_n\}$ in S and

a sequence $\{f_n\}$ in $\mathcal{F}$

Such that

$$\lim_{n\to\infty} (x_n - y_n) = 0 \quad \text{and}$$

$$|f_n(x_n) - f_n(y_n)| \geq \varepsilon,$$

$n \in \mathbb{N}$.

(2) If $\mathcal{F}$ is equicontinuous, then each $f \in \mathcal{F}$ is uniformly continuous, but the converse is false.

For example, $f_n : [0,1] \longrightarrow \mathbb{R}$

$$f_n(x) = x^n, \; n \in \mathbb{N}$$

$0 \leq x \leq 1$.

Each f_n is uniformly continuous.

$$\left.\begin{aligned} x_n &= 1 \\ y_n &= \left(\tfrac{1}{2}\right)^{1/n} \end{aligned}\right\} \text{ for each } n \in \mathbb{N}$$

$$|x_n - y_n| = \left|1 - \left(\tfrac{1}{2}\right)^{1/n}\right|$$

$$\longrightarrow 0 \quad , \text{as } n \to \infty$$

and

$$|f_n(x_n) - f_n(y_n)|$$

$$= \left|1 - \tfrac{1}{2}\right|$$

$$= \tfrac{1}{2} \quad , \forall n \in \mathbb{N}$$

$\Rightarrow \mathcal{F} = \{ f_n : n \in \mathbb{N} \}$ is

NOT equicontinuous

family.

Theorem [118] Let S be a compact in $\mathbb{R}$ and $\{f_n : n \in \mathbb{N}\}$ be a sequence of continuous functions from S to $\mathbb{R}$. Suppose that $f : S \to \mathbb{R}$ is continuous and $\{f_n\} \to f$ pointwise.

① [Dini's theorem] If either $f_1 \leq f_2 \leq \cdots$ (or) $f_1 \geq f_2 \geq f_3 \geq \cdots$ then $\{f_n\} \to f$ uniformly.

② If $\{f_n : n \in \mathbb{N}\}$ is equicontinuous then $\{f_n\} \to f$ uniformly.

<u>Proof</u>

① Assume $f_1 \le f_2 \le f_3 \le \cdots$

and put $g_n := f - f_n$.

Then $g_1 \geq g_2 \geq g_3 \geq \cdots \geq 0$

and $\{g_n\} \longrightarrow 0$ pointwise.

Assume that $\{g_n\} \longrightarrow 0$ uniformly

$\exists\, \varepsilon > 0$, a strictly increasing

sequence $\{n_k\}$ in $\mathbb{N}$ and

a sequence $\{x_k\}$ in S s.t.

$$g_{n_k}(x_k) \geq \varepsilon \quad , \forall\, k \in \mathbb{N}.$$

Since S is compact, the sequence $\{x_k\}$ has a convergent subsequence (again denote it by x_k)

$x_k \longrightarrow x$ for some $x \in S$.

For any $m \geq k$ we have

$$g_{n_k}(x_m) \geq g_{n_m}(x_m) \geq \varepsilon.$$

Since $\{x_k\} \longrightarrow x$ and g_{n_k} is continuous map, we have

$$g_{n_k}(x) \geq \varepsilon \text{ for every } k \in \mathbb{N}.$$

This contradicts the fact that

$\{g_n\} \longrightarrow 0$ pointwise.

② Given $\{f_n : n \in \mathbb{N}\}$ is a equicontinuous family.

Since f is uniformly continuous, by taking $f_0 = f$, we see that $\{f_n : n \geq 0\}$ is equicontinuous family.

Given $\varepsilon > 0$, choose $\delta > 0$ s.t

$$f_n((x-\delta, x+\delta)) \subset \left(f_n(x) - \frac{\varepsilon}{3}, f_n(x) + \frac{\varepsilon}{3}\right)$$

for every $n \geq 0$.

Since S is compact, there exists points $x_1, x_2, x_3, \ldots x_k \in S$ s.t $S = \bigcup_{j=1}^{k} (x_j - \delta, x_j + \delta)$, Since $\{f_n\} \longrightarrow f$ pointwise

$\exists N_\varepsilon \in \mathbb{N}$ s.t

$$|f_n(x_j) - f(x_j)| < \varepsilon/3$$

for $n \geq N_\varepsilon$, $1 \leq j \leq k$.

Now consider $x \in S$ and choose $j \in \{1, 2, 3, \dots, k\}$ with $x \in (x_j - \delta, x_j + \delta)$

Then for any $n \geq N_\varepsilon$, we have

$$|f(x) - f_n(x)|$$

$$\leq |f(x) - f(x_j)| + |f(x_j) - f_n(x_j)| + |f_n(x_j) - f_n(x)|$$

$$< \varepsilon/3 + \varepsilon/3 + \varepsilon/3$$

$$= \varepsilon.$$

This shows that $\{f_n\} \to f$ uniformly.

Series of real functions

Definition: Let $S \subseteq \mathbb{R}$ and $f_n : S \to \mathbb{R}$ be functions $(n \in \mathbb{N})$

Let $s_n : S \to \mathbb{R}$ be

$$s_n := f_1 + f_2 + f_3 + \cdots + f_n.$$

① We say the series $\sum_{n=1}^{\infty} f_n$ converges at $x \in S$ if the real sequence $\sum_{n=1}^{\infty} f_n(x)$ is convergent.

Equivalently, if the sequence $\{s_n(x)\}$ converges.

(2) We say $\sum_{n=1}^{\infty} f_n$ is absolutely convergent at $x \in S$ if

$$\sum_{n=1}^{\infty} |f_n(x)| < \infty.$$

(3) We say $\sum_{n=1}^{\infty} f_n$ converges uniformly to a function $f : S \longrightarrow \mathbb{R}$ if $\{s_n\} \longrightarrow f$ uniformly on S.

(4) We say $\sum_{n=1}^{\infty} f_n$ is uniformly Cauchy if the sequence $\{s_n\}$

is uniformly Cauchy

i.e., for every $\varepsilon > 0$ $\exists$ $N_\varepsilon \in \mathbb{N}$

such that

$$\sup\{|S_n(x) - S_m(x)| : x \in S\} < \varepsilon, \quad \forall\ n, m \geq N_\varepsilon.$$

Examples:

① $f_n : \mathbb{R} \longrightarrow \mathbb{R}$ given by

$$f_n(x) = \frac{x^2}{n^2}$$

$$\sum_{n=1}^{\infty} |f_n(x)| = \sum_{n=1}^{\infty} \frac{x^2}{n^2}$$

$$= x^2 \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$$

for each $x \in \mathbb{R}$.

It follows that $\sum_{n=1}^{\infty} f_n$ converges

absolutely at each $x \in \mathbb{R}$

and hence $\sum_{n=1}^{\infty} f_n$ converges at

each $x \in \mathbb{R}$.

However, we show that the

convergence is NOT uniform on $\mathbb{R}$.

By defining

$$f(x) = \sum_{n=1}^{\infty} f_n(x) ,$$

we see that

$$| f(n+1) - s_n(n+1) |$$

$$\geqslant | f_{n+1}(n+1) |$$

$= 1 \quad , \quad \forall\, n \in \mathbb{N}.$

② Let $f_n : (0, \infty) \longrightarrow \mathbb{R}$ be defined by

$$f_n(x) = x^{\log(n)} \, ,$$

$0 \leq x < \infty.$

Here

$$x^{\log(n)} = \left(e^{\log(x)}\right)^{\log(n)}$$

$$= \left(e^{\log(n)}\right)^{\log(x)}$$

$$= n^{\log(x)}$$

$$\sum_{n=1}^{\infty} f_n(x) = \sum_{n=1}^{\infty} n^{\log(x)}$$

$$= \sum_{n=1}^{\infty} \frac{1}{n^{-\log(x)}}$$

This implies that

$\sum_{n=1}^{\infty} f_n(x)$ converges at

$x \in (0, \infty)$

$$\Longleftrightarrow$$

$\sum_{n=1}^{\infty} \frac{1}{n^{-\log(x)}}$ Converges

$$\Longleftrightarrow$$

$$-\log(x) > 1$$

$\iff \quad x < \frac{1}{e}.$

Hence $\sum_{n=1}^{\infty} f_n$ converges at every $x \in (0, 1/e)$.

Theorem [119]

(Weierstrass M-test)

Let $S \subseteq \mathbb{R}$ and $f_n : S \longrightarrow \mathbb{R}$ be functions for $n \in \mathbb{N}$.

Suppose there exists $K \in \mathbb{N}$ and $M_n \geq 0$ for $n \in \mathbb{N}$ s.t $\sum_{n=1}^{\infty} M_n < \infty$

and $|f_n(x)| \leq M_n$ for every $n \geq K$, $x \in S$

then the series

$$\sum_{n=1}^{\infty} f_n \text{ converges absolutely}$$

and uniformly on S.

<u>Proof.</u> Given that there is $k \in \mathbb{N}$ and $M_n \geq 0$ for $n \in \mathbb{N}$ with

$$\sum_{n=1}^{\infty} M_n < \infty \quad \text{and} \quad |f_n(x)| \leq M_n$$

for every $n \geq k$, $x \in S$.

It is clear that

$$\sum_{n=1}^{\infty} |f_n(x)| \leq \sum_{n=1}^{k} |f_n(x)| + \sum_{n=k+1}^{\infty} |f_n(x)|$$

$$\leq \sum_{n=1}^{k} |f_n(x)| + \sum_{n=k+1}^{\infty} M_n$$

$$< \infty ,$$

for each $x \in S$.

This means that $\sum_{n=1}^{\infty} f_n$ Converges absolutely and so, Converges.

Let us define $f : S \longrightarrow \mathbb{R}$ as,

$$f(x) = \sum_{n=1}^{\infty} f_n(x).$$

To show uniform Convergence of the series, we must show that $\{S_n\} \longrightarrow f$ uniformly where $s_n(x) = f_1(x) + f_2(x) + \cdots + f_n(x)$

For $\varepsilon > 0$, choose $N_\varepsilon > k$ with

$$\sum_{n=N_\varepsilon}^{\infty} M_n < \varepsilon \text{ , then}$$

for any $n \geq N_\varepsilon$ and $x \in S$

we have

$$|f(x) - s_n(x)| \leq \sum_{m=n+1}^{\infty} |f_m(y)|$$

$$\leq \sum_{m=n+1}^{\infty} M_m$$

$$< \varepsilon$$

Thus $\{s_n\} \longrightarrow f$ uniformly on S.

Example:

Let $f_n : [0,\infty) \longrightarrow \mathbb{R}$ be

given by

$$f_n(x) = \frac{x^n}{e^{nx}} \; ; \; 0 \le x < \infty$$

If $x \in [0, \frac{1}{2}]$ then

$$\frac{x}{e^x} \le \frac{1}{2}$$

If $x \in [\frac{1}{2}, 3]$ then

$$\frac{x}{e^x} \le \frac{x}{1+x} = \frac{1}{1+\frac{1}{x}}$$

$$\le \frac{1}{1+1/3} = \frac{3}{4}$$

If $x \in [3, \infty)$ then

$$\frac{x}{e^x} \leq \frac{x}{x^2/2} = \frac{2}{x} \leq \frac{2}{3}$$

Now consider $\max\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}\right\}$

$$= \frac{3}{4}$$

and so, take $M_n = \left(\frac{3}{4}\right)^n$

$\sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n < \infty$ and

$f_n(x) \leq \left(\frac{3}{4}\right)^n \quad \forall\, x \in [0, \infty)$
$n \in \mathbb{N}$

By Weierstrass M-test, we have $\sum_{n=1}^{\infty} f_n$ converges uniformly on $[0, \infty)$

MTH 205: ANALYSIS IN ONE VARIABLE

Lecture 13 DATE : 17.03.2025

DIFFERENTIATION

Let $U \subseteq \mathbb{R}$ be an open set and

$f : U \longrightarrow \mathbb{R}$ be a function.

(i) We say that f is differentiable at $a \in U$ if

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

exists as a real number.

i.e., there is some $L \in \mathbb{R}$ with the property that

$\forall \varepsilon > 0 \ \exists \ \delta > 0$ such that

$(a-\delta, a+\delta) \subset U$ and

$$\left| \frac{f(x)-f(a)}{x-a} - L \right| < \varepsilon$$

$$\Rightarrow \quad |f(x)-f(a)-L(x-a)| < \varepsilon(x-a),$$

$$\forall \; x \in (a-\delta, a+\delta) \setminus \{a\}.$$

Equivalently,

f is differentiable at 'a' if

$$\lim_{h \to 0} \frac{f(a+h)-f(a)}{h} = L$$

for some $L \in \mathbb{R}$.

For $\varepsilon > 0$ $\exists$ $\delta > 0$ s.t $(a-\delta, a+\delta) \subset U$

and

$$|f(a+h)-f(a)-Lh| < \varepsilon |h|$$

$\forall\ h \in \mathbb{R}$ with $0 < |h| < \delta$.

(ii) If f is differentiable at $a \in U$ then the derivative $f'(a)$ of f at 'a' is defined as.

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

$$= \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

(iii) We say that 'f' is differentiable in U if 'f' is differentiable at each point of U.

Note: $f: U \to \mathbb{R}$ is differentiable at $a \in U$ with $f'(a) = L$

$$\iff \lim_{n\to\infty} \frac{f(x_n) - f(a)}{x_n - a} = L$$

for every sequence $\{x_n\}$ in $U \setminus \{a\}$ that converges to a.

$\boxed{\text{Remark}}$

1. Geometrically, $\dfrac{f(x) - f(a)}{x - a}$ is the slope of the line joining $(a, f(a))$ and $(x, f(x))$.

Hence $f'(a)$ is the slope of the tangent line to the graph of f at $(a, f(a))$.

② we may think of $\dfrac{f(x)-f(a)}{x-a}$

as a quantity measuring the rate of change of f near 'a'.

$f'(a)$ - rate of change of 'f' at 'a'.

If we thing of $f(x)$ as the position of an object at time x, then $f'(a)$ is the velocity of the object at time 'a' and the second derivative $f''(a) := (f')'(a)$

is the acceleration of the object at time 'a'.

③ For a function $f:[a,b] \longrightarrow \mathbb{R}$ the differentiability of 'f' at the end points 'a' and 'b' are defined by considering the one-sided limits

$$\lim_{x \to a^+} \frac{f(x)-f(a)}{x-a} \quad \text{and} \quad \lim_{x \to b^-} \frac{f(x)-f(b)}{x-b}$$

<u>Eg:</u> Define $f: \mathbb{R} \longrightarrow \mathbb{R}$ by

$$f(x) = |x-2|, \ \forall x \in \mathbb{R}.$$

Here 'f' is continuous but f is NOT differentiable at '2'.

This is because

$$\lim_{n\to\infty} \frac{f\left(2+\frac{1}{n}\right)-f(2)}{\left(2+\frac{1}{n}\right)-2}$$

$$= 1.$$

but

$$\lim_{n\to\infty} \frac{f\left(2-\frac{1}{n}\right)-f(2)}{\left(2-\frac{1}{n}\right)-2}$$

$$= -1.$$

Indeed (why?) f is differentiable

in $\mathbb{R}\setminus\{2\}$.

More generally, if $\alpha_1, \alpha_2, \cdots \alpha_n \in \mathbb{R}$

are finitely many points

then the continuous function

$f : \mathbb{R} \longrightarrow \mathbb{R}$ given by

$$f(x) = \sum_{j=1}^{n} |x - \alpha_j| \text{ is}$$

differentiable in $\mathbb{R} \setminus \{\alpha_1, \ldots \alpha_n\}$.

<u>Theorem [120]</u>

Let $U \subset \mathbb{R}$ be open, $f : U \longrightarrow \mathbb{R}$ be a function and $a \in U$. Then

① [caratheodory lemma]

f is differentiable at a

$\Longleftrightarrow$ there is a function

$f_a : U \longrightarrow \mathbb{R}$ such that

$$f(x) - f(a) = (x-a)\, f_a(x),$$
$$\forall\, x \in U$$

and f_a is continuous at 'a'.

Here f_a may not be continuous at other points of U. If such a function f_a exists, then it is necessarily unique and is called the <u>Caratheodory function</u> of f at 'a'.

(2) If f is differentiable at 'a' then f is continuous at 'a'.

(3) If f is differentiable at 'a' and f is non-vanishing

in a neighborhood of 'a',

then $1/f$ is differentiable

at a and

$$\left(1/f\right)'(a) = \frac{-f'(a)}{(f(a))^2}.$$

<u>Proof</u>:

(1) Suppose that f is differentiable at $a \in U$.

i.e.,

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists.

Define $f_a : U \longrightarrow \mathbb{R}$ by

$$f_a(x) = \begin{cases} \dfrac{f(x) - f(a)}{x - a}, & \text{if } x \neq a \\ f'(a) & \text{if } x \neq a \end{cases}$$

Clearly, f_a is a continuous function at 'a'; and

$$f(x) - f(a) = (x - a) f_a(x), \quad \forall x \in U.$$

Now conversely, assume that such a function f_a exists.

This means that

$$f_a(x) = \frac{f(x)-f(a)}{x-a},$$

$$x \in U \setminus \{a\}.$$

Since f_a is continuous at 'a' we get

$$\lim_{x \to a} \frac{f(x)-f(a)}{x-a} = f_a(a)$$

$$\Rightarrow \quad f'(a) = f_a(a)$$

$\Rightarrow$ f is differentiable at $a \in U$.

This proof also shows why f_a is unique.

② Suppose that f is differentiable at $a \in U$.

Let f_a be the Caratheodory function for f at a.

If $\{x_n\} \longrightarrow a$ in U

then $\{f_a(x_n)\} \longrightarrow f_a(a)$

by continuity of f_a at 'a'.

Hence

$f(x_n) - f(a)$

$$= (x_n - a) f_a(x_n)$$

$$\longrightarrow 0 \quad \text{as } n \to \infty.$$

f is continuous at 'a'.

(3) Now assume that f is differentiable at 'a' and f is non-vanishing in a neighborhood of a.

Let f_a be the Caratheodory function for f at a, then

$$\frac{(1/f)(x) - (1/f)(a)}{x-a}$$

$$= \frac{f(a) - f(x)}{(x-a) f(x) f(a)}$$

$$= \frac{-(x-a)\ f_a(x)}{(x-a)\ f(x) f(a)}$$

$$= \frac{-\ f_a(x)}{f(x)\ f(a)}$$

Since f is continuous at 'a'

and $f_a(a) = f'(a)$, we have

$$\lim_{x \to a} \frac{(1/f)(x) - (1/f)(a)}{x-a}$$

$$= \lim_{x \to a} - \frac{f_a(x)}{f(x)\, f(a)}$$

$$= \frac{-f'(a)}{[f(a)]^2}.$$

It follows that

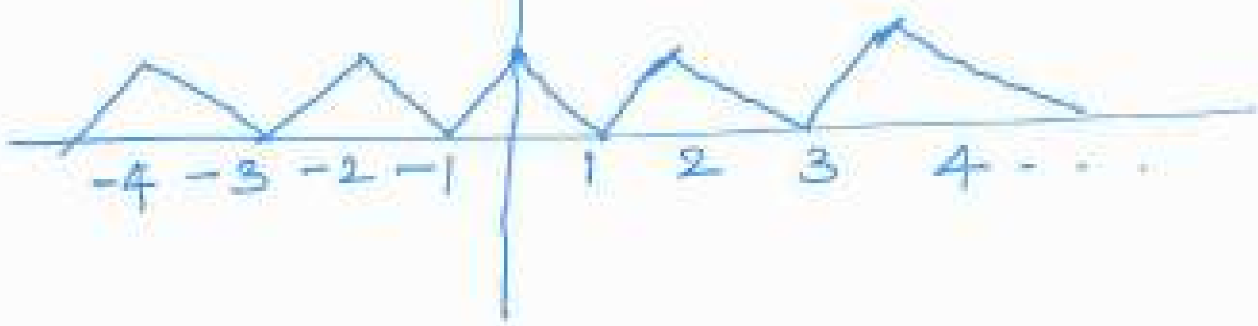
$$(1/f)'(a) = \frac{-f'(a)}{[f(a)]^2}$$

Eg: There exists a Continuous function f on $\mathbb{R}$ which is nowhere differentiable

Define $\varphi(x) = |x|$, on $[-1,1]$

Now extend the definition of $\varphi(x)$ to all of $\mathbb{R}$ such that

$$\varphi(x+2) = \varphi(x) \quad \forall x \in \mathbb{R}$$



then for all $x, a \in \mathbb{R}$

$$|\varphi(x) - \varphi(a)| \leq |x-a|$$

In particular, φ is continuous on $\mathbb{R}$

Define $f : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$f(x) = \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \varphi(4^n x),$$

$x \in \mathbb{R}$

By the Weierstrass M-test the above series converges uniformly on $\mathbb{R}$ and so, f is continuous on $\mathbb{R}$

Now we show that f is not differentiable at any real number.

Let $x \in \mathbb{R}$. For each positive integer $m \in \mathbb{Z}$, put

$$\delta_m = \pm \frac{1}{2} 4^{-m},$$

where the sign is choosen that no integer lies between $4^m x$ and $4^m(x+\delta_m)$.

This is possible because

$$4^m |\delta_m| = \frac{1}{2}.$$

Define

$$t_n := \frac{\varphi(4^n(x+\delta_m)) - \varphi(4^n x)}{\delta_m}$$

when $n>m$ then

$$4^n \delta_m \text{ is an even integer.}$$

So that $t_n = 0$.

when $0 \le n \le m$. we have

$$|t_n| \le 4^n.$$

Since $|t_m| = 4^m$, it

follows that

$$\left| \frac{f(x+\delta_m) - f(x)}{\delta_m} \right|$$

$$= \left| \sum_{n=0}^{m} \left(\frac{3}{4}\right)^n t_n \right|$$

$$\geq 3^m - \sum_{n=0}^{m-1} 3^n$$

$$= \frac{1}{2}\left(3^m + 1\right).$$

As $m \to \infty$, $\delta_m \to 0$

but the above limit does not exist

Hence f is not differentiable at any $x \in \mathbb{R}$.

Lecture 14 DATE : 23.03.2025

Recall that $f : U \longrightarrow \mathbb{R}$ is differentiable at $a \in U$ if and only if there a carathéodory function $f_a : U \longrightarrow \mathbb{R}$ such that

- f_a is continuous at a
- $f(x) - f(a) = (x-a) f_a(x)$, $\forall x \in U$.
- The derivative of f at 'a' is given by $f'(a) = f_a(a)$.

Remark

Let $f : U \longrightarrow \mathbb{R}$ be differentiable at $a \in U$. Then

$$\lim_{h \to 0} \frac{f(a+h) - f(a-h)}{2h}$$

$$= \lim_{h \to 0} \frac{f(a+h) - f(a)}{2h}$$

$$+ \lim_{h \to 0} \frac{f(a-h) - f(a)}{-2h}$$

$$= \frac{1}{2} f'(a) + \frac{1}{2} f'(a)$$

$$= f'(a)$$

If $f : U \longrightarrow \mathbb{R}$ is differentiable at $a \in U$, then

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a-h)}{2h}$$

On the other hand, the 'limit'

$$\lim_{h \to 0} \frac{f(a+h) - f(a-h)}{2h} \text{ may exist}$$

without f being differentiable.

For example,

$f : \mathbb{R} \longrightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

f is not continuous at $x=0$.

but

$$\lim_{h\to 0} \frac{f(h)-f(-h)}{2h} = 0.$$

Note: Let $f: \mathbb{R} \to \mathbb{R}$ be differentiable and suppose for every $\varepsilon > 0$ $\exists\ \delta > 0$ such that

$$\left| \frac{f(a+h)-f(a)}{h} - f'(a) \right| < \varepsilon,$$

for every $a \in \mathbb{R}$ and every $0 < |h| < \delta$.

Define $g_n : \mathbb{R} \to \mathbb{R}$ by

$$g_n(x) = \frac{f(x+1/n)-f(x)}{1/n},$$

$\forall\ x \in \mathbb{R}$.

Then $\{g_n\}$ is a sequence of continuous functions.

Let $\varepsilon>0$. Now choose $N\in\mathbb{N}$ s.t $\frac{1}{N}<\delta$. From the hypothesis

$$|g_n(x)-f'(x)|<\varepsilon \quad , \text{ for } n\geq N \text{ and } x\in\mathbb{R}$$

This shows that

$$\{g_n\}\longrightarrow f' \text{ uniformly.}$$

Hence f' is continuous.

<u>Theorem [121]</u> Let $U\subseteq\mathbb{R}$ be open and $f,g:U\longrightarrow\mathbb{R}$ be differentiable at $a\in U$. Then

① $\alpha \cdot f$, $f+g$ are differentiable at 'a' with

$$(\alpha \cdot f)'(a) = \alpha \cdot f'(a) \quad \text{and}$$

$$(f+g)'(a) = f'(a) + g'(a),$$

② fg is differentiable at 'a' with

$$(fg)'(a) = f'(a)\,g(a) + f(a)\,g'(a)$$

③ If g is non-vanishing in U then f/g is differentiable at 'a' with

$$(f/g)'(a) = \frac{g(a)\,f'(a) - f(a)\,g'(a)}{[g(a)]^2}.$$

Proof: (i) Let $\alpha \in \mathbb{R}$ and $f, g : U \longrightarrow \mathbb{R}$ differentiable at $a \in U$. Then

$$\lim_{x \to a} \frac{(\alpha \cdot f + g)(x) - (\alpha \cdot f + g)(a)}{x - a}$$

$$= \alpha \lim_{x \to a} \frac{f(x) - f(a)}{x - a} + \lim_{x \to a} \frac{g(x) - g(a)}{x - a}$$

$$= \alpha \cdot f'(a) + g'(a).$$

(2) Let f_a and g_a represent Caratheodory function of

f and g at 'a' respectively.

We see that

$$f(x)\,g(x) - f(a)\,g(a)$$

$$= \left[f(x)\,g(x) - f(a)\,g(x)\right]$$

$$+ \left[f(a)\,g(x) - f(a)\,g(a)\right]$$

$$= \left[f(x) - f(a)\right] g(x)$$

$$+ \; f(a)\left[g(x) - g(a)\right]$$

$$= (x-a)\,f_a(x)\,g(x)$$

$$+ \; f(a)\,(x-a)\,g_a(x)$$

By using the continuity of f_a, g, g_a at 'a'

we get

$$\lim_{x \to a} \frac{f(x)\,g(x) - f(a)\,g(a)}{x-a}$$

$$= f_a(a)\,g(a) + f(a)\,g_a(a)$$

$$= f'(a)\,g(a) + f(a)\,g'(a).$$

(3) Here g is non-vanishing in U.

We know from Theorem [120] that $1/g$ is differentiable and the derivative of $(1/g)$ at 'a' is given by

$$\left(1/g\right)'(a) = \frac{-g'(a)}{[g(a)]^2}$$

Now applying (2) we see that

(f/g) is differentiable at a

and

$$\left(f/g\right)'(a) = f'(a)\left(1/g\right)(a) + f(a)\left(1/g\right)'(a)$$

$$= \frac{f'(a)}{g(a)} - \frac{f(a)\,g'(a)}{[g(a)]^2}$$

$$= \frac{g(a)\,f'(a) - f(a)\,g'(a)}{[g(a)]^2}.$$

Example: For $n = 0, 1, 2, 3, \ldots$

define $f_n : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$f_n(x) = x^n, \quad \forall x \in \mathbb{R}$$

In particular, $f_0 \equiv 1$, $f_1(x) = x$, $x \in \mathbb{R}$

and $f_0' \equiv 0$, $f_1' \equiv 1$.

Since $f_2(x) = f_1(x) f_1(x)$,

by product rule,

$$f_2'(x) = f_1'(x) f_1(x) + f_1(x) f_1'(x)$$

$$= 2x$$

Inductively,

$$f_n'(x) = n \cdot x^n \quad , n \in \mathbb{N}.$$

If $f : \mathbb{R} \longrightarrow \mathbb{R}$ is a non-constant polynomial (say)

$$f(x) = \sum_{k=0}^{n} a_k x^k$$

then 'f' is differentiable in $\mathbb{R}$

and $$f'(x) = \sum_{k=1}^{n} k a_k x^{k-1}.$$

Here $f'(x)$ is again a polynomial with $\deg(f') = \deg(f) - 1$.

As a result,

$\frac{d^n}{dx^n} f(x) = f^{(n)}(x)$ is a constant and

$$f^{(m)}(x) = 0 \quad \text{for every } m \geq n+1.$$

Theorem [122] Chain rule

Let $U, V \subseteq \mathbb{R}$ be open, $f: U \to V$ and $g: V \to \mathbb{R}$ be a function.

If f is differentiable at $a \in U$ and g is differentiable at $f(a)$ then $g \circ f : U \to \mathbb{R}$ is differentiable at 'a' and

$$(g \circ f)'(a) = g'(f(a)) \cdot f'(a).$$

Proof: Let f_a and $g_{f(a)}$ be Caratheodory functions of f at 'a' and g at $f(a)$ respectively.

So, we can write

$$g(f(x)) - g(f(a))$$

$$= [f(x) - f(a)]\, g_{f(a)}(f(x))$$

Since f is differentiable at 'a'

and $g_{f(a)}(f(a)) = g'(f(a))$,

it follows that

$$\lim_{x \to a} \frac{g(f(x)) - g(f(a))}{x - a}$$

$$= \lim_{x \to a} \frac{f(x) - f(a)}{x - a}\, g_{f(a)}(f(a))$$

$$= f'(a)\, g'(f(a)).$$

This shows that $(g \circ f)$ is differentiable at 'a' and

$$(g \circ f)'(a) = g'(f(a)) \cdot f'(a).$$

Eg: (1) $\frac{d}{dx} \sin(x) = \cos(x)$,

$$\frac{d}{dx} \cos(x) = -\sin(x),$$

$$\frac{d}{dx} e^x = e^x.$$

Moreover, if $f : \mathbb{R} \longrightarrow \mathbb{R}$ is differentiable in $\mathbb{R}$ then

$$\frac{d}{dx}\left(e^{f(x)}\right) = e^{f(x)} \cdot f'(x)$$

Note that

$$\sin(x) - \sin(y)$$

$$= 2\cos\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)$$

and

$$\lim_{x\to 0}\frac{\sin(x)}{x} = 1.$$

So, $\displaystyle\lim_{h\to 0}\frac{\sin(x+h)-\sin(x)}{h}$

$$= \lim_{h\to 0}\frac{2\cos\left(x+\frac{h}{2}\right)\sin\left(\frac{h}{2}\right)}{h}$$

$$= \left[\lim_{h\to 0}\cos\left(x+\frac{h}{2}\right)\right]$$

$$\left[\lim_{h\to 0}\frac{\sin(h/2)}{h/2}\right]$$

$$= \cos(x)$$

$$\boxed{\frac{d}{dx}\sin(x) = \cos(x).}$$

similarly,

$$\lim_{h\to 0} \frac{\cos(x+h)-\cos(x)}{h}$$

$$= \lim_{h\to 0} \frac{-2\sin\left(x+h/2\right)\sin\left(h/2\right)}{h}$$

$$= \left[\lim_{h\to 0} -\sin\left(x+h/2\right)\right]$$

$$\left[\lim_{h\to 0} \frac{\sin\left(h/2\right)}{h/2}\right]$$

$$= -\sin(x)$$

$$\boxed{\frac{d}{dx}\cos(x) = -\sin(x)}$$

$$\lim_{h\to 0} \frac{e^{x+h} - e^{x}}{h}$$

$$= \lim_{h\to 0} \frac{e^{x}\cdot e^{h} - e^{x}}{h}$$

$$= e^{x} \lim_{h\to 0} \frac{e^{h} - 1}{h}$$

$$= e^{x} \lim_{h\to 0} \left[1 + \frac{h}{2!} + \frac{h^2}{3!} + \cdots\right]$$

$$\boxed{\frac{d}{dx} e^{x} = e^{x}.}$$

Suppose that f is differentiable in $\mathbb{R}$, and take $g(x) = e^x$

then $$(g \circ f)(x) = e^{f(x)}$$

Hence by chain rule,

$$\frac{d}{dx} e^{f(x)} = g'(f(x))\, f'(x)$$
$$= e^{f(x)} \cdot f'(x).$$

$$\boxed{\frac{d}{dx} e^{f(x)} = e^{f(x)} \cdot f'(x)}$$

Example:

Let $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} x\sin(1/x) & , x \neq 0 \\ 0 & \text{o.w} \end{cases}$$

$$g(x) = \begin{cases} x^2\sin(1/x) & , x \neq 0 \\ 0 & . \text{o.w} \end{cases}$$

We know that f, g are continuous on $\mathbb{R}$.

Also note that

$$\lim_{h \to 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \to 0} \frac{h \sin(1/h)}{h}$$

$$= \lim_{h \to 0} \sin(1/h) \quad \text{does not exist}$$

This shows that $f(x)$ is not differentiable at $x=0$.

Since $1/x$ is differentiable at $x \neq 0$ by the chain rule and product rule $f(x) = x \sin(1/x)$ is differentiable at $x \in \mathbb{R} \setminus \{0\}$ and

$$\boxed{f'(x) = \sin(1/x) - \frac{1}{x} \cos(1/x)}$$

Also, g is differentiable on $\mathbb{R}$ with

$$\boxed{g'(x) = 2x \sin(1/x) - \cos(1/x)}$$

In particular,

$$g'(0) = \lim_{h\to 0} \frac{g(h)-g(0)}{h}$$

$$= \lim_{h\to 0} \frac{h^2 \sin(1/x)}{h}$$

$$= \lim_{h\to 0} h \sin(1/h)$$

$$= 0.$$

$$g'(x) = \begin{cases} 2x\sin(1/x) - \cos(1/x), & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Now consider the sequence

$$\left\{ \frac{1}{2n\pi} : n \geq 1 \right\} \text{ in } \mathbb{R}$$

Here $\frac{1}{2n\pi} \longrightarrow 0$, as $n \to \infty$

$$g'\left(\frac{1}{2n\pi}\right)$$

$$= 2\left(\frac{1}{2n\pi}\right)\sin(2n\pi) - \cos(2n\pi)$$

$$= -\cos(2n\pi)$$

$$\longrightarrow 1 \quad \text{as } n \to \infty.$$

$$\neq g'(0).$$

Hence g' is not continuous at $x = 0$.

MTH 205: ANALYSIS IN ONE VARIABLE

Lecture 15 DATE : 24.03.2025

Definition :

Let $X \subseteq \mathbb{R}$, $f : X \longrightarrow \mathbb{R}$ be a function and $x_0 \in X$. We say that f has a local minimum (respectively, local maximum) at x_0 if there is an open neighborhood U of x_0 in X s.t

$$f(x) \geq f(x_0)$$

$$\left(\text{respectively, } f(x) \leq f(x_0)\right)$$

for every $x \in U$.

Theorem [123]

① If $f : (a,b) \longrightarrow \mathbb{R}$ is a function having either a local minimum (or) a local maximum at a point $c \in (a,b)$ and if f is differentiable at c, then $f'(c) = 0$.

② [Darboux theorem - IVP of derivative]

Let $f : [a,b] \longrightarrow \mathbb{R}$ be differentiable. If either $f'(a) < y < f'(b)$

$$(\text{or}) \quad f'(a) > y > f'(b)$$

then there is $c \in (a,b)$ with $f'(c) = y$.

Proof.

① Assume that f has local minimum at $c \in (a,b)$. Let $\varepsilon > 0$ be such that $(c-\varepsilon, c+\varepsilon) \subset (a,b)$ and

$$f(x) \geq f(c) \;, \; \forall \; x \in (c-\varepsilon, c+\varepsilon).$$

Then for $0 < h < \varepsilon$, we have

$$f(c \pm h) \geq f(c) \quad \text{and so,}$$

$$\frac{f(c-h) - f(c)}{-h} \leq 0 \leq \frac{f(c+h) - f(c)}{h}.$$

$$\Rightarrow \lim_{h \to 0} \frac{f(c-h) - f(c)}{-h} \leq 0 \leq \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

$$\Rightarrow \boxed{f'(c) = 0}.$$

(2) Assume that

$$f'(a) < y < f'(b).$$

Define $g : [a,b] \longrightarrow \mathbb{R}$ as

$$g(x) = f(x) - yx ,$$
$$a \le x \le b$$

Then g is differentiable

and $g'(a) < 0 < g'(b)$

Since g is continuous and $[a,b]$ is compact, g attains its minimum at some $c \in [a,b]$.

Since $\lim_{h \to 0^+} \frac{g(a+h) - g(a)}{h} = g'(a) < 0$

and $\lim_{h \to 0^+} \frac{g(b-h) - g(b)}{h} = g'(b) > 0$

$\exists \varepsilon > 0$ s.t

$g(a+h) < g(a)$ and $g(b-h) < g(b)$

for $0 < h < \varepsilon$.

Consequently, $c \notin \{a, b\}$.

Therefore, $c \in (a,b)$ and so

by 1, $g'(c) = 0$.

It follows that

$$\boxed{f'(c) = y}$$

<u>Remark</u> 1 Let $f : [1,2] \longrightarrow \mathbb{R}$

be defined by

$$f(x) = x^2, \quad 1 \leq x \leq 2.$$

Then f attains its minimum at $x=1$ and maximum at $x=2$.

However, $f'(1) = 2 \neq 0$

and $f'(2) = 4 \neq 0$.

THE MEAN VALUE THEOREM

Theorem [124]

① [Rolle's theorem] Let $f:[a,b] \to \mathbb{R}$ be continuous. If f is differentiable in (a,b) and $f(a) = f(b)$ then there is $c \in (a,b)$ with $f'(c) = 0$

② Let $f, g : [a,b] \longrightarrow \mathbb{R}$ be continuous and differentiable in (a,b). Then there is $c \in (a,b)$ with

$$[f(b) - f(a)]\, g'(c) = [g(b) - g(a)]\, f'(c).$$

③ (Mean value theorem)

Let $f : [a,b] \longrightarrow \mathbb{R}$ be continuous and assume f is differentiable in (a,b). Then there is $c \in (a,b)$ with

$$f(b) - f(a) = (b-a)\, f'(c).$$

Proof. ①

Since f is continuous function and $[a,b]$ is compact, f attains its minimum and maximum in $[a,b]$.

Let $u = \min_{a \le x \le b} f(x)$ and $v = \max_{a \le x \le b} f(x)$.

If $u = v$, then f is constant and hence $f' \equiv 0$ in (a,b).

If $u < v$ then atleast one of them must differ from $f(a) = f(b)$.

Suppose $u < f(a) = f(b)$, then

$f(c) = u$ for some $c \in (a,b)$.

Here f attains minimum at 'c'

and so, $\boxed{f'(c) = 0}$

(2) Let $h : [a,b] \longrightarrow \mathbb{R}$ be

defined by

$$h(x) = [f(b) - f(a)]\, g(x) - [g(b) - g(a)]\, f(x)$$

then 'h' is continuous on $[a,b]$

and differentiable in (a,b).

Moreover,

$$h(a) = f(b)g(a) - g(b)f(a)$$
$$= h(b).$$

Hence by Rolle's theorem, there is $c \in (a,b)$ with $h'(c)=0$.

It follows that

$$[f(b)-f(a)]\, g'(c) = [g(b)-g(a)]\, f'(c).$$

3 Now consider $g(x)=x$ on $[a,b]$

Here f, g are continuous on $[a,b]$ and differentiable in (a,b).

From 2 it follows that there is $c \in (a,b)$ such that

$$[f(b)-f(a)] = (b-a)\, f'(c).$$

Theorem [125]

① Let $f : [a,b] \longrightarrow \mathbb{R}$ be differentiable with $f' \equiv 0$. Then f is constant.

② Let $f, g : [a,b] \longrightarrow \mathbb{R}$ be differentiable with $f' = g'$ then $f - g$ is constant.

③ Let $f : [a,b] \longrightarrow \mathbb{R}$ be differentiable. Then

f is increasing (or decreasing)

if and only if

$$f' \geqslant 0 \quad (\text{or } f' \leq 0).$$

④ If $f : [a,b] \longrightarrow \mathbb{R}$ is a C^1-map (i.e., if f is diff. with f' continuous) then

f is Lipschitz continuous on $[a,b]$. In particular, every polynomial $f : [a,b] \longrightarrow \mathbb{R}$ is Lipschitz continuous on $[a,b]$.

Proof. ① Let $x, y \in [a,b]$ with $x<y$. By the mean value theorem there is $c \in (x,y)$ s.t

$$f(y) - f(x) = (y-x) f'(c).$$

Since $f'(c) = 0$, we get

$$f(y) = f(x).$$

$\therefore$ f is a constant function.

② Here $f' = g'$ i.e.,

$$(f-g)' \equiv 0$$

By apply ① to $(f-g)$ we conclude the result.

(3) Suppose f is increasing and $c \in [a,b]$. Then for every $x \in [a,b] \setminus \{c\}$ we have

$$\frac{f(x)-f(c)}{x-c} \geq 0$$

By letting $x \to c$ we get

$$f'(c) \geq 0.$$

Conversely, assume that $f' \geq 0$.

For $x < y$ in $[a,b]$, $\exists$ $c \in (x,y)$

s.t $f(y)-f(x) = f'(c)(y-x)$

Since $f' \geqslant 0$ and $x \leq y$ we have

$$f(y) \geqslant f(x).$$

Thus f is increasing.

(4) Since f' is continuous and $[a,b]$ is compact, there is $\lambda > 0$ s.t

$$|f'(c)| \leq \lambda, \forall c \in [a,b]$$

Now if $x<y$ in $[a,b]$ then there is $c \in (x,y)$ with

$$f(y) - f(x) = (y-x) f'(c)$$

by mean value theorem and

hence

$$|f(x) - f(y)| \leq \lambda |x - y|.$$

Thus f is Lipschitz Continuous.

Lecture 16 DATE : 25.03.2025

Recall: The mean value theorem states that

if $f:[a,b]\longrightarrow\mathbb{R}$ continuous and differentiable in (a,b).

Then there is a $c\in(a,b)$

such that

$$f(b)-f(a)=(b-a)f'(c).$$

Example:

(1) $\tan(x) > x$, for $0 < x < \pi/2$.

Let $f: [0, \pi/2) \to \mathbb{R}$ be given by

$$f(x) = \tan(x).$$

Note that

$$f'(x) = \frac{d}{dx}\left(\frac{\sin(x)}{\cos(x)}\right)$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x}$$

Hence for any $x \in (0, \pi/2)$ there is $y \in (0, x)$ by mean value theorem

$$f(x)-f(0) = x f'(y)$$

i.e.,

$$\boxed{\tan(x) = x\frac{1}{\cos^2 y} > x.}$$

Hence the result.

② Let $f:\mathbb{R}\to\mathbb{R}$ be differentiable with $f'=f$ and $f(0)=1$.

Then $f(x)=e^x$, $\forall x\in\mathbb{R}$.

First consider the function

$$g(x) = f(x)\,e^{-x}.$$

By product rule,

$$g'(x) = f'(x)e^{-x} - f(x)e^{-x}$$

$$= 0$$

Thus g is constant function

and so,

$$g(x) = g(0) = 1 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow \quad f(x) = e^{x}, \ \forall x \in \mathbb{R}$$

problems:

① Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be differentiable with $\lim\limits_{x \to \infty} f'(x) = 0$ and

$f(n) = 0 \ \forall n \in \mathbb{N}$ then

Show that

$$\lim_{x \to \infty} f(x) = 0.$$

Solution: Given that $\lim_{x \to \infty} f'(x) = 0$

and $f(n) = 0$, $\forall n \in \mathbb{N}$.

Let $x \in \mathbb{R}$. Then there is $N \in \mathbb{N}$

s.t $N < x \leq N+1$.

f is continuous on the interval

$[N, x]$ and differentiable

in (N, x). By the mean value

theorem $\exists\, c \in (N, x)$ s.t

$$f(x) - f(N) = (x-N) f'(c).$$

$$\Rightarrow \quad f(x) < f'(c)$$

Note that $c \to \infty$ whenever $x \to \infty$

by the choice of interval.

Thus
$$\lim_{n\to\infty} f(x) = 0.$$

(2) Let $f : \mathbb{R} \to \mathbb{R}$ be a C^2-map

with
$$f(1/n) = 1/n^2 \quad \forall n \in \mathbb{N}$$

then show that
$$f''(0) = 2.$$

Solution:

Let $g(x) = f(x) - x^2$ on $\mathbb{R}$

Note that

$$g\left(\frac{1}{n}\right) = 0 = g\left(\frac{1}{n+1}\right)$$

g is continuous on $\left[\frac{1}{n+1}, \frac{1}{n}\right]$

and differentiable in $\left(\frac{1}{n+1}, \frac{1}{n}\right)$

By Rolle's theorem

$$\exists \; c_n \in \left(\frac{1}{n+1}, \frac{1}{n}\right), \quad n \in \mathbb{N}$$

s.t $\quad g'(c_n) = 0.$

Now note that (since g is C^2-map)

g' is continuous on $[c_n, c_{n+1}]$

and differentiable in (c_n, c_{n+1})

with $g'(c_n) = g'(c_{n+1})$

$\exists$ $d_n \in (c_n, c_{n+1})$ s.t

$$g''(d_n) = 0.$$

Since $d_n \to 0$ and g'' is continuous function,

$$g''(0) = 0.$$

Hence $\boxed{f''(0) = 2}$

③ If $f : [0,1] \longrightarrow \mathbb{R}$ is a differentiable map with $f(0)=1$ and $f(1) = 1/e$ then there is $c \in (0,1)$ with

$$f(c) = f'(c).$$

Solution:

Let $g(x) = e^{x} f(x)$

Then $g(0) = 1 = g(1)$

By Rolle's theorem $\exists\, c \in (0,1)$ s.t $g'(c) = 0$

It follows that

$$e^{-c} f'(c) - f(c) e^{-c} = 0.$$

So, $\boxed{f(c) = f'(c)}$

(4) If $f : [1, e] \longrightarrow \mathbb{R}$ is differentiable, then there is $c \in (1, e)$ with

$$f(e) - f(1) = c f'(c).$$

<u>Solution:</u> Let $g(x) = \log(x)$ on $[1, e]$

By generalized mean value theorem. $\exists c \in (1, e)$

$$[f(e) - f(1)]\, g'(c) = [g(e) - g(1)]\, f'(c).$$

Thus

$$[f(e) - f(1)] = c f'(c).$$

(5) If $f:[1,\sqrt{3}] \longrightarrow \mathbb{R}$ is differentiable, then there is $c \in (1,\sqrt{3})$ with

$$f(\sqrt{3})-f(1) = \frac{f'(c)}{c}.$$

Solution: Now consider

$$g(x)=x^2$$

Apply generalized mean value thm, $\exists c \in (1,\sqrt{3})$

$$[f(\sqrt{3})-f(1)]\, g'(c)$$
$$= [g(\sqrt{3})-g(1)]\, f'(c)$$

Therefore,

$$f(\sqrt{3})-f(1) = \frac{f'(c)}{c}.$$

Lecture 17 DATE : 01.04.2025

If f has an n^{th} derivative at a point $x_0 \in U$, then it is possible to construct a polynomial 'P_n' of degree 'n' s.t

$$P_n(x_0) = f(x_0) \quad \text{and}$$

$$P_n^{(k)}(x_0) = f^{(k)}(x_0) \quad \text{for}$$

$$k = 1, 2, 3, \dots n.$$

i.e.,

$$P_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n.$$

This polynomial P_n is called

n^{th} Taylor polynomial for f at x_0.

Now this polynomial P_n give a reasonable approximation to 'f' for points near x_0.

The following result provides such information.

"Taylor's Theorem".

$\boxed{\text{Taylor's Theorem}}$ Let $n \in \mathbb{N}$ and let $f : [a,b] \longrightarrow \mathbb{R}$ be continuous such that

$f^{(k)}$ is continuous on $[a,b]$ $(1 \leq k \leq n)$

and $f^{(n)}$ is differentiable on (a,b)

If $x_0 \in [a,b]$ then for any $x \in [a,b]$ there exists a point c between x_0 and x satisfying,

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + \frac{f^{(n+1)}(c)}{(n+1)!}(x-x_0)^{n+1}$$

Proof. Let $J = [x_0, x]$ and define $F : J \longrightarrow \mathbb{R}$ by

$$F(t) := f(x) - f(t) - (x-t) f'(t) - \cdots - \frac{f^{(n)}(t)}{n!} (x-t)^n,$$

$\forall \, t \in J.$

Here F is differentiable and

$$F'(t) = - \frac{(x-t)^n}{n!} f^{(n+1)}(t).$$

If we define $G : J \longrightarrow \mathbb{R}$ by

$$G(t) := F(t) - \left(\frac{x-t}{x-x_0}\right)^{n+1} F(x_0),$$

for $t \in J$.

then

$$G(x_0) = F(x_0) - F(x_0)$$
$$= 0$$

and

$$G(x) = F(x)$$
$$= 0$$

By applying Rolle's theorem, there exist $c \in (x_0, x)$ s.t

$$G'(c) = 0.$$

This implies that

$$F'(c) + (n+1)\frac{(x-c)^n}{(x-x_0)^{n+1}}F(x_0) = 0.$$

We obtain that

$$F(x_0) = -\frac{1}{n+1}\frac{(x-x_0)^{n+1}}{(x-c)^n}F'(c)$$

$$= \frac{1}{n+1}\frac{(x-x_0)^{n+1}}{(x-c)^n}\frac{(x-c)^n}{n!}f^{(n+1)}(c)$$

$$= \frac{f^{(n+1)}(c)}{(n+1)!}(x-x_0)^{n+1}$$

As a result,

$$f(x) - f(x_0) - (x-x_0)f'(x_0)$$

$$- \cdots - \frac{(x-x_0)^n}{n!} f^{(n)}(x_0)$$

$$= \frac{f^{(n+1)}(c)}{(n+1)!}(x-x_0)^{n+1}$$

Hence

$$f(x) = f(x_0) + (x-x_0)f'(x_0)$$

$$+ \frac{(x-x_0)^2}{2!} f''(x_0) +$$

$$\cdots + \frac{(x-x_0)^n}{n!} f^{(n)}(x_0)$$

$$+ \frac{(x-x_0)^{n+1}}{(n+1)!} f^{(n+1)}(c).$$

Note: We use the notation

$P_n(x)$ — n^{th} Taylor polynomial of 'f'.

$$R_n(x) := \frac{(x-x_0)^{n+1}}{(n+1)!} f^{(n+1)}(c).$$

Remainder function.

This formula is also referred to as the "Lagrange form"

(or)

the "derivation form".

Example

① [Approximate value of $\sqrt{2}$]

Let $f(x) = x^{1/2}$ for $x > 0$.

Then $f'(x) = \dfrac{x^{-1/2}}{2}$ and

$$f''(x) = \frac{-x^{-3/2}}{4}.$$

Applying Taylor's theorem

with $x_0 = 1$, $x = 2$

and $n = 1$,

then $\exists\ c \in (1,2)$ with

$$f(x) = f(x_0) + (x-x_0) f'(1) + \frac{(x-x_0)^2}{2!} f''(c).$$

i.e

$$\sqrt{2} = 1 + (2-1)\frac{1}{2} + \frac{(2-1)^2}{2!}\left(\frac{-c^{-3/2}}{4}\right)$$

$$= \frac{3}{2} - \frac{c^{-3/2}}{8}.$$

Since $c \in (1,2)$, we have

$$-1 \leq -c^{-3/2} \leq -2^{-3/2} \leq \frac{-1}{3}$$

Hence

$$\frac{3}{2} - \frac{1}{8} \leq \sqrt{2} \leq \frac{3}{2} - \frac{1}{24}$$

② [Approximate value of e]

Let $f(x) = e^x$. Applying Taylor's theorem with $x_0 = 0$, $x = 1$ and $n = 3$, then we find $c \in (0,1)$ s.t

$$f(x) = f(x_0) + (x - x_0) f'(x_0) + \frac{(x-x_0)^2}{2!} f''(x_0) + \frac{(x-x_0)^3}{3!} f^{(3)}(x_0) + \frac{(x-x_0)^4}{4!} f^{(4)}(c)$$

i.e.

$$e = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} e^c$$

$$= \frac{8}{3} + \frac{e^c}{24}$$

Since

$$1 = e^0 \le e^c \le e^1 \le 3$$

we deduce that

$$\frac{8}{3} + \frac{1}{24} \le \frac{8}{3} + \frac{e^c}{24} \le \frac{8}{3} + \frac{3}{24}$$

Hence

$$2.7 \le e \le 2.8.$$

(3) Use Taylor's theorem with $n=2$ to approximate $(1+x)^{1/3}$, for $x>-1$.

Soln: We take the function

$$f(x) := (1+x)^{1/3}$$ and

the point $x_0 = 0$, $n=2$.

Here $f'(x) = \frac{1}{3}(1+x)^{-2/3}$

and

$$f''(x) = -\frac{2}{9}(1+x)^{-5/3}$$

By Taylor's theorem,

$$f(x) = f(x_0) + (x-x_0) f'(x_0) + \frac{(x-x_0)^2}{2!} f''(x_0) + R_2(x),$$

with

$$R_2(x) = \frac{(x-x_0)^3}{3!} f^{(3)}(x_0).$$

for some $c \in (0, x)$.

Since $x_0 = 0$, we have

$$f(x) = \underbrace{1 + \frac{1}{3}x - \frac{1}{9}x^2}_{P_2(x)} + R_2(x).$$

with

$$R_2(x) = \frac{1}{3!} x^3 f^{(3)}(c)$$

$$= \frac{1}{3!} x^3 \left[\frac{10}{27}(1+c)^{-8/3}\right]$$

$$= \frac{5}{81}(1+c)^{-8/3} x^3$$

if $x = 0.3$ we get the

approximation

$$P_2(0.3) = 1.09 \text{ for } (1.3)^{1/3}$$

and since $c \in (0, 0.3)$

we see that

$$(1+c)^{-8/3} < 1 .$$

So the error is almost

$$R_2(0.3)$$

$$= \frac{5}{81}(1+c)^{-8/3}(0.3)^3$$

$$\leq \frac{5}{81}(0.3)^3$$

$$= \frac{1}{600}$$

$$< 0.17 \times 10^{-2}$$

Hence we have

$$\left|(1.3)^{1/3} - 1.09\right|$$

$$= \left|f(0.3) - P_2(0.3)\right|$$

$$= \left| R_2(0.3) \right|$$

$$< 0.17 \times 10^{-2}$$

Two decimal places accuracy

is assured.

Lecture 18 DATE : 04.04.2025

Theorem [125] [Taylor Series]

(i) Let $f : [a,b] \longrightarrow \mathbb{R}$ be a C^{∞}-map (an infinitely differentiable map). Assume that

$$|f^{(n)}(x)| \leq M \quad , \forall x \in [a,b], \; n \in \mathbb{N}$$

Then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n,$$

for every $x \in [a,b]$

Here the series converges to f

uniformly.

(ii) $\sin(x)$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7} + \cdots$$

and

$\cos(x)$

$$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

for every $x \in \mathbb{R}$.

The convergence of two series are uniform on $[-r, r]$, for each $r > 0$.

(iii) Let $a \in \mathbb{R}$, $r > 0$ and

$f : (a-r, a+r) \longrightarrow \mathbb{R}$ be a C^{∞}-map.

Assume $\exists M > 0$ s.t

$$|f^{(n)}(x)| \leq M \frac{n!}{r^n},$$

for every $x \in (a-r, a+r)$, $n \in \mathbb{N}$.

Then $$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n,$$

for $x \in (a-r, a+r)$ and the series converges to f uniformly in $[a-t, a+t]$, $\forall\ t \in (0, r)$.

Proof.

(i) Note that the n^{th} partial sum of the series is,

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k, \quad n \in \mathbb{N}.$$

Clearly,

$$P_n(a) = f(a) \quad \text{for every } n \in \mathbb{N}$$

Now if $x \in (a,b]$ then by Taylor's theorem $\exists\, c \in (a,x)$ such that

$$f(x) = P_n(x) + f^{(n+1)}(c)\frac{(x-a)^{n+1}}{(n+1)!}$$

Hence

$$|f(x) - P_n(x)|$$

$$= |f^{(n+1)}(c)| \frac{(x-a)^{n+1}}{(n+1)!}$$

$$\leq M \frac{(b-a)^{n+1}}{(n+1)!}$$

$$\longrightarrow 0, \text{ as } n \to \infty$$

(independent of 'x')

Thus $\{P_n\} \longrightarrow f$ uniformly

and so,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

(ii) Fix $r>0$ and take

$$f(x) = \sin(x) \quad , \ x \in [-r, r]$$

Here f is a C^{∞}-map and

$$|f^{(n)}(x)| \leq 1 \quad \text{for every } x \in [-r,r], \ n \in \mathbb{N}$$

In particular, if we consider $x \in [0,r]$ and applying (i) we get

$$\sin(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots$$

$$+ \frac{f^n(0)}{n!} x^n + \cdots$$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$$

$$+ \cdots$$

On the other hand, if $x \in [-r, 0]$

then $-x \in [0, r]$ and so

$$\sin(-x) = (-x) - \frac{(-x)^3}{3!} +$$

$$\frac{(-x)^5}{5!} - \frac{(-x)^7}{7!} + \cdots$$

This implies

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$\forall\, x \in [-r, r]$

As $r \in \mathbb{R}$ is choosen arbitrarily, the result holds true.

Similarly,

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

(iii) Let $a \in \mathbb{R}$ and $r > 0$;

$f : (a-r, a+r) \longrightarrow \mathbb{R}$ be a C^∞-map.

Let $t \in (0, r)$. We will show that the series $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$ converges uniformly to f on

$[a, a+t]$ and similar argument works for $[a-t, a]$.

Note that the n^{th} partial sum is $P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$.

and $P_n(a) = f(a)$, $\forall n \in \mathbb{N}$.

Now, if $x \in (a, a+t]$ then by Taylor's theorem $\exists\, c \in (a, x)$ such that

$$f(x) = P_n(x) + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

Thus

$$|f(x) - P_n(x)| \leq \frac{M\, t^{n+1}}{r^{n+1}}$$

$$\longrightarrow 0, \text{ as } n \to \infty$$

independent of x because

$0 < t < r$.

Hence the result.

Remark:

The series expansion of $\sin(x)$, $\cos(x)$ given above are called the Taylor series of $\sin(x)$

Problems

① If $f:[a,b] \longrightarrow \mathbb{R}$ is a function with $f^{(n+1)} \equiv 0$, then f is a polynomial of degree $\leq n$.

Soln: Fix $x \in (a,b]$ then

$\exists\, c \in (a,x)$ s.t

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

$$+ \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

Since $f^{(n+1)} \equiv 0$ we have

$$f(x) = P_n(x) \quad \text{for } x \in [a,b]$$

② Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be twice differentiable with

$$M := \sup\{|f''(x)| : x \in \mathbb{R}\} < \infty.$$

and

$$\lim_{x \to \infty} f(x) = 0 \quad \text{then}$$

Show that $\lim_{x \to \infty} f'(x) = 0.$

<u>Hint</u>: Given that f'' exists on $\mathbb{R}$.

For any $a<b$, by Taylor's thm,

$\exists\, c \in (a,b)$ s.t

$$f(b) = f(a) + f'(a)(b-a) + \frac{f''(c)}{2!}(b-a)^2.$$

It follows that

$$f'(a) = \frac{f(b)-f(a)}{b-a} - \frac{f''(c)}{2}(b-a)$$

so,

$$|f'(a)| \leq \frac{|f(b)-f(a)|}{b-a} + \frac{|f''(c)|(b-a)}{2}$$

$$\leq \frac{|f(b)-f(a)|}{b-a} + \frac{M(b-a)}{2}$$

Complete the solution.

(3) Let $f : [0,1] \longrightarrow \mathbb{R}$ be infinitly differentiable. Assume $f(1/n) = 0$ for every $n \in \mathbb{N}$ and

$$M = \sup\left\{ |f^{(n)}(x)| : x \in [0,1] \ \& \ n \in \mathbb{N} \right\}$$

then

$$f \equiv 0 \text{ on } [0,1].$$

Sol^n: Since $f(1/n) = 0$ for each $n \in \mathbb{N}$

and f is continuous

$$f(0) = 0.$$

For each $n \in \mathbb{N}$, by the mean value theorem $\exists\, c_n \in \left(\frac{1}{n+1}, \frac{1}{n}\right)$ s.t

$$f'(c_n) = 0.$$

Note that the sequence $\{c_n\}$ converges to '0' as $n \to \infty$ and by continuity of "f'" it follows that

$$f'(0) = 0.$$

In general $f^{(k)}(0) = 0$, for $k \in \mathbb{N} \cup \{0\}$

Now by Taylor's theorem,

for each $b \in (0,1]$ there is

$c \in (0,b)$ with

$$f(b) = 0 + \frac{f^{n+1}(c)}{(n+1)!}(b-0)$$

and hence

$$|f(b)| \leq \frac{Mb}{(n+1)!} \quad \forall n \in \mathbb{N}.$$

Therefore, $f(b) = 0$

and the result follows.

④ Show that

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

for $|x| < 1/2$

and the series converges uniformly on $[-t, t]$ for every $t \in (0, 1/2)$

MTH 205: ANALYSIS IN ONE VARIABLE

Lecture 19

DATE : 15.04.2025

Riemann Integral

Let $I = [a,b]$ be a closed interval in $\mathbb{R}$.

A partition of I is a finite, ordered set

$$P := \{x_0, x_1, x_2, x_3, \ldots, x_n\}$$

of points in I such that

$$a = x_0 < x_1 < x_2 < x_3 < \cdots < x_n = b.$$

The points of P are used to devide $I = [a,b]$ into non-overlaping Sub intervals

$$I_1 = [x_0, x_1] \quad I_2 = [x_1, x_2]$$

$$\cdots \quad I_n = [x_{n-1}, x_n]$$

- The norm (or) mesh of P is the number

$$\|P\| = \max_{1 \leq i \leq n} (x_i - x_{i-1})$$

- The norm of partition is merely the length of the largest Subinterval into which the partition devides $[a,b]$

- If $t_i \in [x_{i-1}, x_i]$ $\quad 1 \leq i \leq n$

then the points are called 'tags' of sub interval $[x_{i-1}, x_i]$

- A set of ordered pairs

$$\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$$

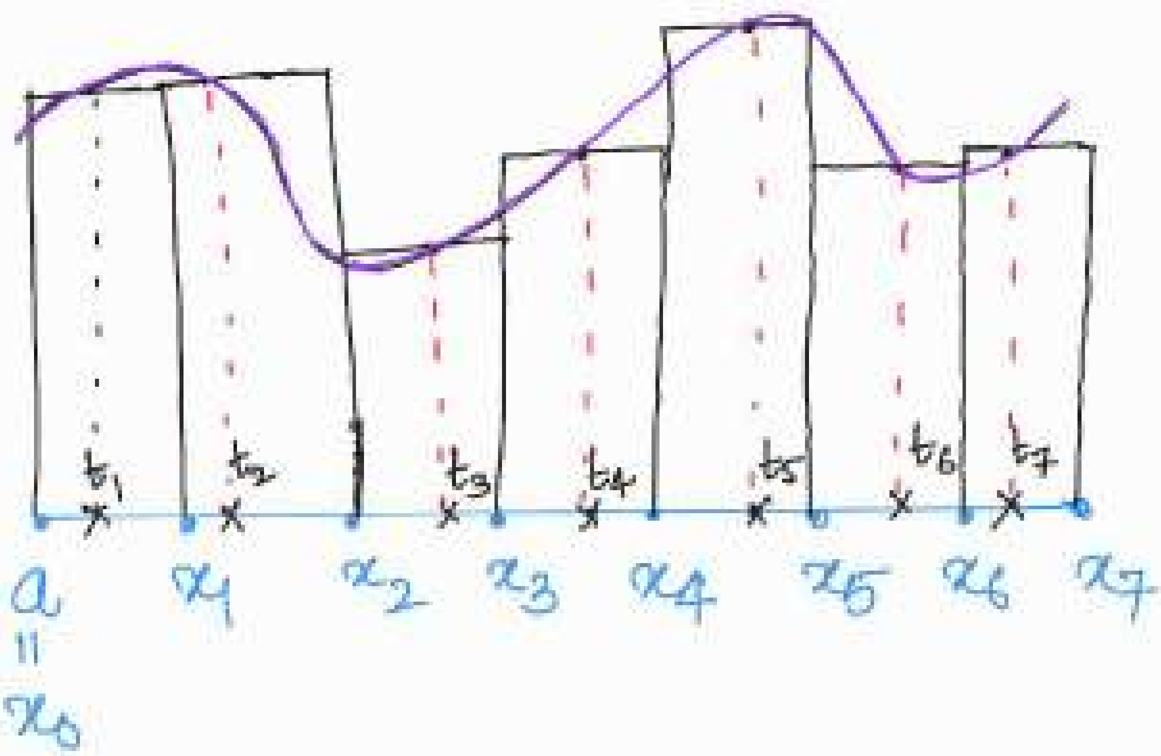
of subintervals and corresponding tags is called tagged partition of I.

- If $\dot{P}$ is the tagged partition, the <u>Riemann Sum</u> of a function $f : [a,b] \longrightarrow \mathbb{R}$ corresponding to $\dot{P}$ is defined as

$$S(f;\dot{P}) = \sum_{i=1}^{n} f(t_i)\,(x_i - x_{i-1})$$

- Let us understand "Riemann sum" with an example. If f is a positive function on $[a,b]$, the Riemann sum is the sum of the areas of n rectangles whose bases are subintervals $[x_{i-1}, x_i]$ and whose heights are $f(t_i)$

See the Figure 1 below.



Definition:

A function $f : [a,b] \longrightarrow \mathbb{R}$ is said to be "Riemann Integrable" on $[a,b]$ if there is a number $L \in \mathbb{R}$ s.t $\forall \varepsilon > 0 \ \exists \ \delta > 0$ such that

$$|S(f;\dot{P}) - L| < \varepsilon$$

for any tagged partition $\dot{P}$ with $\|\dot{P}\| < \delta$.

- The set of all Riemann Integrable function is denoted by $R[a,b]$
- Now we show that such L (when exists) is unique.

Theorem [126]

If $f \in R[a,b]$ then the value L is uniquely determined.

Proof. Assume that L_1 and L_2

Satisfy the definite of Riemann Integerable.

For $k \in \mathbb{N}$ $\exists \delta_1 > 0$ s.t

$$\left| S(f, \dot{P}) - L_1 \right| < \frac{1}{2k}$$

for every tagged partition $\dot{P}$ with $\|\dot{P}\| < \delta_1$.

Similarly, $\exists \delta_2 > 0$ s.t

$$\left| S(f ; \dot{P}) - L_2 \right) < \frac{1}{2k}$$

for every tagged partition $\dot{P}$ with $\|\dot{P}\| < \delta_2$.

Now let $\delta = \min\{\delta_1, \delta_2\}$ and if $\dot{P}$ is a tagged partition with $\|\dot{P}\| < \delta$ then

$$|S(f;\dot{P}) - L_1| < \frac{1}{2k} \text{ and}$$

$$|S(f;\dot{P}) - L_2| < \frac{1}{2k}$$

It follows that

$$|L_1 - L_2| \leq |L_1 - S(f;\dot{P})| + |L_2 - S(f,\dot{P})| < \frac{1}{k},$$

for every $k \in \mathbb{N}$.

Thus $\boxed{L_1 = L_2}$

The unique number L is called the "Riemann Integral" of f over $[a,b]$.

It is denoted by

$$L = \int_a^b f(x)\,dx.$$

Properties: Suppose that $f, g \in R[a,b]$ then

① $$\int_a^b (\alpha f)(x)\,dx = \alpha \int_a^b f(x)\,dx$$

for $\alpha \in \mathbb{R}$

② The function $f+g \in R[a,b]$

and $\int_a^b (f+g)(x)\, dx$

$$= \int_a^b f(x)dx + \int_a^b g(x)dx$$

③ If $f(x) \leq g(x), \forall x \in [a,b]$

then $\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx.$

Theorem [127]

If $f \in R[a,b]$ then f is bounded.

__Proof.__ Given that $f \in \mathcal{R}[a,b]$

Assume that f is unbounded function with integral L.

There is $\delta > 0$ s.t

$|S(f;\dot{P}) - L| < 1$ $\left(\Rightarrow |S(f;\dot{P})| < 1 + |L| \right)$ $\circledast$

for any tagged partition $\dot{P}$ with $\|\dot{P}\| < \delta$.

Let $Q = \{ x_0 < x_1 < x_2 < \cdots < x_n \}$ be a partition of $[a,b]$ with $\|Q\| < \delta$.

Since we assume f is unbounded,

f is unbounded on

$[x_{k-1}, x_k]$ for some k

We tag $\dot{Q}$ by taking

$t_i = x_i$ $(i \neq k)$ and choose t_k

s.t

$$\left| f(t_k)\,(x_k - x_{k-1}) \right|$$

$$> 1 + |L| + \left| \sum_{i \neq k} f(t_i)\,(x_i - x_{i-1}) \right|$$

then

$$|S(f;\dot{Q})|$$

$$\geq |f(t_k)(x_k - x_{k-1})|$$

$$- \left|\sum_{i \neq k} f(t_i)(x_i - x_{i-1})\right|$$

$$> 1 + |L|$$

This is a contradiction to $*$

$\therefore$ f is bounded.

Exercise: Consider Tomae's function

$h : [0,1] \longrightarrow \mathbb{R}$ defined as

$$h(x) = \begin{cases} 0 & \text{if } x \text{ is irrational} \\ 1 & \text{if } x = 0 \\ \frac{1}{n} & \text{if } x = \frac{m}{n}, \\ & \gcd(m,n) = 1. \end{cases}$$

Show that 'h' is Riemann integrable.

Solution: Here we show that Tomae's function is Riemann Integrable and $\int_0^1 h(x)\,dx = 0$.

Let $\varepsilon > 0$, then the set

$$E_\varepsilon = \{ x \in [0,1] \,/\, h(x) \geq \varepsilon/2 \}$$

is a finite set. Suppose that there are N_ε number of elements.

Let $\delta = \varepsilon / 4N_\varepsilon$ and $\dot{P}$ is a tagged partition with

$\|\dot{P}\| < \delta$.

If $\dot{P}_1$ is the subset of

$\dot{P}$ having tags in E_ε

and $\dot{P}_2$ is the subset of

$\dot{P}$ having tags elsewhere in $[0,1]$

Note that $\dot{P}_1$ has atmost $2N_\varepsilon$

intervals whose total length is

less than $2N_\varepsilon \delta = \varepsilon/2$ and

$0 < h(t_i) \leq 1$ for every tag in $\dot{P}_1$.

Also the total lengths of the

subintervals in $\dot{P}_2$ is ≤ 1

and $h(t_i) < \varepsilon/2$ for every

tag in $\dot{P}_2$. Therefore, we have

$$
\begin{aligned}
|S(f;\dot{P})| &= |S(f;\dot{P}_1) + S(f;\dot{P}_2)| \\
&< 1 \cdot 2N_\varepsilon \delta + \varepsilon/2 \cdot 1 \\
&= \varepsilon/2 + \varepsilon/2 \\
&= \varepsilon.
\end{aligned}
$$

Since $\varepsilon > 0$ is arbitrary, we infer that $h \in \mathcal{R}[0,1]$ with integral '0'.

Lecture 20 DATE : 21.04.2025

Cauchy criterion

A function $f : [a,b] \longrightarrow \mathbb{R}$ belongs to $\mathcal{R}[a,b]$ if and only if for $\varepsilon > 0$ there exists $\eta_\varepsilon > 0$ such that if $\dot{P}$ and $\dot{Q}$ are any tagged partitions $[a,b]$ with $\|\dot{P}\| < \eta_\varepsilon$ and $\|\dot{Q}\| < \eta_\varepsilon$ then

$$|S(f;\dot{P}) - S(f;\dot{Q})| < \varepsilon.$$

The Squeeze theorem

Let $f : [a,b] \longrightarrow \mathbb{R}$. Then

$f \in \mathcal{R}[a,b]$ if and only if for every $\varepsilon > 0$ $\exists$ functions α_ε and ω_ε in $\mathcal{R}[a,b]$ with

$$\alpha_\varepsilon(x) \leq f(x) \leq \omega_\varepsilon(x)$$

for all $x \in [a,b]$.

and such that

$$\int_a^b (\omega_\varepsilon - \alpha_\varepsilon)(x)\, dx < \varepsilon .$$

<u>Proof.</u> $(:\Rightarrow)$ If $f \in \mathcal{R}[a,b]$ then take $\alpha_\varepsilon = \omega_\varepsilon = f$, $\forall\ \varepsilon > 0$.

($\Leftarrow$:) For $\varepsilon > 0$, assume there exists two functions $\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b]$ with

$$\alpha_\varepsilon(x) \leq f(x) \leq \omega_\varepsilon(x), \quad \forall x \in [a,b] \quad \text{— (I)}$$

and $$\int_a^b (\omega_\varepsilon - \alpha_\varepsilon)(x)\, dx < \varepsilon. \quad \text{— (II)}$$

Since $\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b]$ there is $\delta > 0$ s.t for any tagged partition $\dot{P}$ with $\|\dot{P}\| < \delta$ we have

$$\left| S(\alpha_\varepsilon; \dot{P}) - \int_a^b \alpha_\varepsilon \right| < \varepsilon$$

and

$$\left| S(\omega_\varepsilon; \dot{P}) - \int_a^b \omega_\varepsilon \right| < \varepsilon.$$

It follows that

$$\int_a^b \alpha_\varepsilon - \varepsilon < S(\alpha_\varepsilon ; \dot{P})$$

and

$$S(\omega_\varepsilon ; \dot{P}) < \int_a^b \omega_\varepsilon + \varepsilon .$$

From the inequality I, we have

$$S(\alpha_\varepsilon ; \dot{P}) \leq S(f ; \dot{P}) \leq S(\omega_\varepsilon ; \dot{P})$$

and so,

$$\boxed{\int_a^b \alpha_\varepsilon - \varepsilon < S(f ; \dot{P}) < \int_a^b \omega_\varepsilon + \varepsilon} \quad \text{— III}$$

If $\dot{Q}$ is another tagged partition with $\|\dot{Q}\| < \delta$ then we have (similarly)

$$\boxed{\int_a^b \alpha_\varepsilon - \varepsilon < S(f;\dot{Q}) < \int_a^b \omega_\varepsilon + \varepsilon} \quad \text{— IV}$$

Now subtracting III and IV, by using II, we conclude that

$$|S(f;\dot{P}) - S(f;\dot{Q})| < \int_a^b \omega_\varepsilon - \int_a^b \alpha_\varepsilon + 2\varepsilon$$

$$= \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) + 2\varepsilon$$

$$< 3\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, from the Cauchy criterion implies that $f \in R[a,b]$.

Classes of Riemann Integrable Functions

Lemma: If $J = [c,d]$ is a sub interval of $[a,b]$ and

$$\varphi_J(x) = \begin{cases} 1 & \text{if } x \in J \\ 0 & \text{if } x \notin J \end{cases}$$

then $\varphi_J \in R[a,b]$ and

$$\int_a^b \varphi_J(x)\,dx = d-c.$$

Proof (exercise)

Note: If $\varphi : [a,b] \longrightarrow \mathbb{R}$ is a step function, then $\varphi \in R[a,b]$

Theorem [128]

If $f : [a,b] \longrightarrow \mathbb{R}$ is continuous on $[a,b]$ then $f \in R[a,b]$.

Proof. Since $[a,b]$ is compact f is uniformly continuous.

Therefore, given $\varepsilon > 0$ $\exists$ $\delta > 0$

s.t $u, v \in [a,b]$ with

$|u-v| < \delta$ then $|f(u) - f(v)|$
$< \varepsilon/(b-a)$

Let $P = \{ I_i \}_{i=1}^{n}$ be a partition

such that $\|P\| < \delta$ and

let $u_i \in I_i$ be a point

where f attains its minimum

value on I_i. Also, let

$v_i \in I_i$ be a point where

f attains maximum value

on I_i

Say $I_i = [x_{i-1}, x_i]$

Let $\alpha_\varepsilon : [a,b] \longrightarrow \mathbb{R}$ be defined by

$$\alpha_\varepsilon(x) = f(u_i),$$

for $x \in [x_{i-1}, x_i)$

$i = 1, 2, \cdots n-1$

and

$$\alpha_\varepsilon(x) = f(u_n)$$

for $x \in [x_{n-1}, x_n]$

Similarly, define

$\omega_\varepsilon : [a,b] \longrightarrow \mathbb{R}$ by

$$\omega_\varepsilon(x) = f(v_i),$$

for $x \in [x_{i-1}, x_i)$

$i = 1 \cdots n-1$

and

$$\omega_\varepsilon(x) = f(v_n)$$

for $x \in [x_{n-1}, x_n]$

It follows that

$$\alpha_\varepsilon(x) \leq f(x) \leq \omega_\varepsilon(x)$$

$\forall\, x \in [a,b]$.

and

$$0 \leq \int_a^b \left(\omega_\varepsilon(x) - \alpha_\varepsilon(x)\right) dx$$

$$= \sum_{i=1}^{n} \left[f(v_i) - f(u_i)\right] (x_i - x_{i-1})$$

$$< \sum_{i=1}^{n} \frac{\varepsilon}{(b-a)} (x_i - x_{i-1})$$

$$= \varepsilon.$$

Finally, it follows from the Squeeze theorem that $f \in R[a,b]$.

Problems 22.04.2025

① Suppose that $c \le d$ are points in $[a,b]$. If $\varphi : [a,b] \longrightarrow \mathbb{R}$ is given by $(\gamma > 0)$

$$\varphi(x) = \begin{cases} \gamma, & \text{if } x \in [c,d] \\ 0, & \text{elsewhere in } [a,b] \end{cases}$$

Prove that $\varphi \in \mathcal{R}[a,b]$ and $\displaystyle\int_a^b \varphi(x)\,dx = \alpha\,(d-c)$.

Solution:

Let $\varepsilon > 0$. Take $\delta = \dfrac{\varepsilon}{2\gamma}$.

Let $\dot{P}$ is a tagged partition with $\|\dot{P}\| < \delta$. Suppose $J_1, J_2, \ldots J_k$ be subintervals in $\dot{P}$ with tags in $[c, d]$, then

$$[c+\delta, d-\delta] \subset \bigcup_{i=1}^{k} J_i \subset [c-\delta, d+\delta].$$

It follows that

$$\alpha(d-c-2\delta) \le S(\varphi; \dot{P}) \le \gamma(d-c+2\delta).$$

Therefore,

$$\left| S(\varphi ; \dot{P}) - \gamma (d-c) \right|$$
$$< 2\gamma\delta$$
$$= \varepsilon .$$

As a result

$\varphi \in R[a,b]$ and

$$\int_a^b \varphi(x)\, dx$$
$$= \gamma (d-c).$$

(2) Let $\dot{P} = \{(I_i, t_i)\}_{i=1}^{n}$ be a tagged partition of $[a,b]$ and let $c_1 < c_2$.

(a) If $u \in I_k$ with $c_1 \leq t_k \leq c_2$ then show that

$$c_1 - \|\dot{P}\| \leq u \leq c_2 + \|\dot{P}\|.$$

(b) If $v \in [a,b]$ with

$$c_1 + \|\dot{P}\| \leq v \leq c_2 - \|\dot{P}\|$$

then the tag t_i of any sub interval I_i that contains v satisfies $t_i \in [c_1, c_2]$

Proof. (1) If $u \in I_k = [x_{k-1}, x_k]$

then $x_{k-1} \le u$ so that

$$c_1 \le t_k \le x_k \le x_{k-1} + \|\dot{P}\|$$

$$\Rightarrow \quad c_1 - \|\dot{P}\| \le x_{k-1} \le u. \qquad \text{— (1)}$$

Also, $u \le x_i$ so that

$$x_k - \|\dot{P}\| \le x_{k-1} \le t_k \le c_2$$

$$\Rightarrow \quad u \le x_k \le c_2 + \|\dot{P}\| \qquad \text{— (2)}$$

From (1) and (2), we have

$$c_1 - \|\dot{P}\| \le u \le c_2 + \|\dot{P}\|$$

(2) Here $v \in [x_{i-1}, x_i]$ for some i.

If $c_1 + \|\dot{P}\| \leq v \leq x_i$ then

$$c_1 \leq x_i - \|\dot{P}\| \leq x_{i-1}$$

and

if $v \leq c_2 - \|\dot{P}\|$, then

$$x_i \leq x_{i-1} + \|\dot{P}\| \leq c_2.$$

Therefore,

$$c_1 \leq x_{i-1} \leq t_i \leq x_i \leq c_2.$$

(3) If $f \in R[a,b]$ and if $\{\dot{P}_n\}$ is any sequence of tagged partitions of $[a,b]$

such that $\|\dot{P}_n\| \to 0$,

then show that

$$\int_a^b f(x)\,dx = \lim_n S(f;\dot{P}_n)$$

Proof. Given $\varepsilon > 0$ $\exists$ $\delta > 0$ such that if $\|\dot{P}\| < \delta$ then

$$\left| S(f;\dot{P}) - \int_a^b f(x)\,dx \right| < \varepsilon. \quad \text{---} \; (*)$$

Since $\|\dot{P}_n\| \longrightarrow 0$, there is $N_\varepsilon \in \mathbb{N}$ s.t

$$\|\dot{P}_n\| < \delta, \; \forall \, n \geq N_\varepsilon,$$

It follows from equation (*) that

$$\left| S(f; \dot{P}_n) - \int_a^b f(x)\,dx \right| < \varepsilon,$$

for every $n \geq N_\varepsilon$.

Therefore,

$$\int_a^b f(x)\,dx = \lim_{n \to \infty} S(f; \dot{P}_n)$$

④ Let

$$g(x) = \begin{cases} 0 & \text{if } x \in [0,1] \text{ is rational} \\ \frac{1}{x} & \text{if } x \in [0,1] \text{ is irrational.} \end{cases}$$

Explain why $g \notin R[0,1]$.

However, show that there is a sequence $\{\dot{P}_n\}$ of tagged partitions of $[a,b]$ s.t

$\|\dot{P}_n\| \to 0$ and $\lim_n S(f;\dot{P}_n)$ exists.

Solution: Recall that every Riemann

integrable function over an interval $[a,b]$ is bounded.

Since g is not bounded, it is NOT Riemann integrable over $[0,1]$.

We can construct a sequence $\{\dot{P}_n\}_{n\in\mathbb{N}}$ of tagged partition as follows.

For each $n\in\mathbb{N}$, $\dot{P}_n$ be the partition of $[0,1]$ into equal subintervals with tags at the left end points, which are rational number

i.e.,

$$\dot{P}_1 = \{([0,1], 0)\}$$

$$\dot{P}_2 = \left\{\left(\left[0, \tfrac{1}{2}\right], 0\right), \left(\left[\tfrac{1}{2}, 1\right], \tfrac{1}{2}\right)\right\}$$

$$\dot{P}_3 = \left\{\left(\left[0, \tfrac{1}{3}\right], 0\right), \left(\left[\tfrac{1}{3}, \tfrac{2}{3}\right], \tfrac{1}{3}\right), \left(\left[\tfrac{2}{3}, 1\right], \tfrac{2}{3}\right)\right\}.$$

$\vdots$

Clearly, $\|\dot{P}_n\| = \frac{1}{n} \longrightarrow 0$

and $S(g;\dot{P}_n) = 0 \ \forall n$

$\therefore \lim_{n\to\infty} S(g;\dot{P}_n) = 0$

exists.

⑤ Let $f \in \mathcal{R}[a,b]$ and $c \in \mathbb{R}$.

Define $g : [a+c, b+c] \longrightarrow \mathbb{R}$

by $g(y) := f(y-c)$.

Show that $g \in \mathcal{R}[a+c, b+c]$

and that

$$\int_{a+c}^{b+c} g(x)\,dx = \int_a^b f(x)\,dx.$$

Solution: Let $\varepsilon > 0$. There exist $\delta > 0$ such that

if $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$

is a tagged partition of $[a,b]$

with $\|\dot{P}\| < \delta$ then

$$\left| S(f; \dot{P}) - \int_a^b f(x)\,dx \right| < \varepsilon .$$

Then $\dot{Q} = \{([x_{i-1}+c, x_i+c], t_i+c)\}_{i=1}^{n}$

is a tagged partition of

$[a+c, b+c]$ and

$$\|\dot{Q}\| = \|\dot{P}\| < \delta.$$

Moreover,

$$S(g;\dot{Q})$$

$$= \sum_{i=1}^{n} g(t_i+c)(x_i - x_{i-1})$$

$$= \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1})$$

$$= S(f;\dot{P})$$

and so,

$$\left| S(g;\dot{Q}) - \int_a^b f \right| < \varepsilon.$$

$$\therefore \int_{a+c}^{b+c} g(x)\,dx = \int_a^b f(x)\,dx.$$

<u>Note</u>: In the above problem,

if we start with any arbitrary tagged partition

$$\dot{P}_1 = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$$

of $[a+c, b+c]$ then

$$\dot{Q}_1 = \{([x_{i-1}-c, x_i-c], t_i-c)\}_{i=1}^{n}$$

is a tagged partition of $[a, b]$ with

$$\|\dot{P}_1\| = \|\dot{Q}_1\|$$

⑥ Is "Dirchlet function" Riemann integrable over $[0,1]$?

<u>Solution</u>: Define Dirchlet function on $[0,1]$ as

$$f(x) = \begin{cases} 1 & \text{if } x \in [0,1] \text{ is rational} \\ 0 & \text{if } x \in [0,1] \text{ is irrational} \end{cases}$$

Let P_n be a partion of $[0,1]$ into n equal parts.

If $\dot{P}_n$ is the partition with rational tags then

$$S(f;\dot{P}_n) = 1 \quad , \forall n$$

& $\|\dot{P}_n\| \to 0$. as $n \to \infty$.

If $\dot{Q}_n$ is the same partition with irrational tags, then

$$S(f;\dot{Q}_n) = 0$$

& $\|\dot{Q}_n\| \to 0$ as $n \to \infty$.

Hence f is NOT Riemann Integrable over $[0,1]$.

[See problem 3].

use contrapositive statement in problem 3.

(7) Suppose $f : [a,b] \longrightarrow \mathbb{R}$ is a function s.t $f(x) = 0$ except for a finite number of points $c_1, c_2, c_3, \ldots, c_n$ in $[a,b]$. prove that $f \in R[a,b]$ and

$$\int_a^b f(x)\,dx = 0.$$

Solution:

Since $f(c_i) \neq 0$ for $1 \leq i \leq n$

take $M = \max_{1 \leq i \leq n} f(c_i)$

For $\varepsilon > 0$, choose $\delta = \dfrac{\varepsilon}{nM}$.

If $\dot{P} = \{([x_{j-1}, x_j], t_j)\}_{j=1}^{k}$ be a tagged partition with $\|\dot{p}\| < \delta$ then

$$|S(f; \dot{p})|$$

$$= \sum_{j=1}^{k} f(t_j)(x_j - x_{j-1})$$

This value is 0, except when at least one of the taggs t_j is c_i for some i. However,

$$|x_j - x_{j-1}| < \delta = \frac{\varepsilon}{nM} \quad \forall j$$

shows that

$$|S(f; \dot{p})| < \varepsilon$$

$$\therefore \int_a^b f(x)\,dx = 0.$$

(8) If $g \in R[a,b]$ and if $f(x) = g(x)$ except for finite number of points in $[a,b]$ then show that $f \in R[a,b]$ and $\int_a^b f(x)dx = \int_a^b g(x)dx.$

Solution: It follows from problem (7).

(9) If $f, g \in R[a,b]$ and

$f(x) \leq g(x)$ for all $x \in [a,b]$

then $\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx$?

Solution:

Let $\varepsilon > 0$ $\exists$ $\delta > 0$ s.t

if $\dot{p}$ is a tagged partition of $[a,b]$ with $\|\dot{p}\| < \delta$ then

$$\left| S(f;\dot{p}) - \int_a^b f(x)\,dx \right| < \varepsilon/2 \qquad \text{--- (I)}$$

and

$$\left| S(g; \dot{p}) - \int_a^b g(x)\,dx \right| < \varepsilon/2 \quad \text{--- (II)}$$

From equation (I), we can write

$$\int_a^b f(x)\,dx - \varepsilon/2 < S(f; \dot{p}) \quad \text{--- (III)}$$

From equation (II) we can write

$$\int_a^b g(x)\,dx < S(g; \dot{p}) + \varepsilon/2 \quad \text{--- (IV)}$$

Since $f(x) \leq g(x)$ we have

$$S(f; \dot{p}) \leq S(g; \dot{p})$$

It follows from (III) and (IV) that

$$\int_a^b f(x)dx - \varepsilon/2 < \int_a^b g(x)dx + \varepsilon/2$$

$$\Rightarrow \int_a^b f(x)\,dx < \int_a^b g(x)dx + \varepsilon,$$

Since $\varepsilon > 0$ is arbitrary, we get

$$\int_a^b f(x)dx \leq \int_a^b g(x)dx.$$

(10) Define $h : [0,1] \longrightarrow \mathbb{R}$ by

$$h(x) = \begin{cases} x+1 & \text{for } x \in [0,1] \text{ irrational} \\ 0 & \text{for } x \in [0,1] \text{ is irrational} \end{cases}$$

show that $h \notin R[0,1]$.

Solution: Similar to that of problem (6).

(11) Define $u : [0,1] \longrightarrow \mathbb{R}$ by

$$u(x) = \begin{cases} n & \text{if } x = \frac{1}{n}, n \in \mathbb{N} \\ 0 & \text{otherwise.} \end{cases}$$

Is u Riemann integrable?

Solution:

Let P_n be a partition of $[0,1]$ into n equal parts.

$\|P_n\| = \frac{1}{n} \longrightarrow 0$ as $n \to \infty$.

If $\dot{P}_n$ is a tagged partition with same subintervals and $t_1 = 1/n$

then $S(u; \dot{P}_n) = 1 \quad \forall n$.

If $\dot{Q}_n$ is a tagged partition with the same subintervals and tagged by irrational points

then $S(u; \dot{Q}_n) = 0 \quad \forall n$.

$\therefore$ From problem 3 it follows that

$u \notin R[0,1]$.

⑫ Show that a step function

$\varphi : [a,b] \longrightarrow \mathbb{R}$ is

Riemann integrable.

Solution: Let $J = [c,d]$ be

a subinterval of $[a,b]$

$$\varphi_J(x) = \begin{cases} 1 & \text{if } x \in J \\ 0 & \text{if } x \notin J \end{cases}$$

is called an indicator function on J.

Since φ is a step function

it is a linear combinations

of indicator functions

$$\varphi(x) = \sum_{i=1}^{n} r_i \, \varphi_{J_i}(x)$$

for some $r_1, r_2, \cdots r_n \in \mathbb{R}$ and

$J_i = [c_i, d_i]$ subintervals

of $[a, b]$.

From properties of Riemann Integrable functions $\varphi \in R[a,b]$ &

$$\int_a^b \varphi(x)\,dx = \sum_{i=1}^{n} r_i \int_a^b \varphi_{J_i}(x)\,dx$$

$$= \sum_{i=1}^{n} r_i (d_i - c_i)$$

[See problem 1]

(13) If $S(f;\dot{P})$ is any Riemann sum of $f:[a,b]\to\mathbb{R}$ Show that $\exists$ a step function $\varphi:[a,b]\longrightarrow\mathbb{R}$ s.t

$$\int_a^b \varphi(x)\,dx = S(f;\dot{P}).$$

<u>Solution:</u> If $\dot{P}=\{([x_{i-1},x_i],t_i)\}_{i=1}^{n}$ is a tagged partition of $[a,b]$

take

$$\varphi(x)=\begin{cases} f(t_i) & \text{if } x\in[x_{i-1},x_i),\ i=1,\dots,n-1 \\ f(t_n) & \text{if } x\in[x_{n-1},x_n] \end{cases}$$

Here φ is a Step function and so, $\varphi \in R[a,b]$.

Moreover,

$$\int_a^b \varphi(x)\,dx$$

$$= \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1})$$

$$= S(f ; P).$$

(14) Suppose $f : [a,b] \longrightarrow \mathbb{R}$ is continuous and $f(x) \geq 0$ for all $x \in [a,b]$.

prove that if $\int_a^b f(x)\,dx = 0$

then $f(x) = 0$, $\forall x \in [a,b]$.

Solution:

Assume $\exists\, c \in (a,b)$

s.t $f(c) > 0$

Since f is continuous,

for $\varepsilon = \frac{f(c)}{2}$ $\exists\, \delta > 0$ s.t

$|x - c| < \delta \Rightarrow$

$$|f(x) - f(c)| < \frac{f(c)}{2}$$

In particular, $f(x) > \frac{f(c)}{2}$

whenever $x \in (c-\delta, c+\delta)$.

It follows that

$$\int_a^b f(x)\,dx > \int_{c-\delta}^{c+\delta} f(x)\,dx$$

$$> \frac{f(c)}{2}(2\delta)$$

$$= \delta f(c)$$

$$> 0.$$

Same argument holds free end points.

This is a contradiction to the fact that $\int_a^b f(x)dx = 0$

Therefore, $f(x)=0 \ \forall x\in[a,b]$

Remark: The "Continuity" in the hypothesis of problem (14) can not be dropped.

For instance, $f(x)=\begin{cases} 1 & \text{if } x=0 \\ 0 & \text{if } \\ & \text{of } x\in[0,1] \end{cases}$

$$\int_0^1 f(x)\,dx = 0$$

but $f \neq 0$.

Also, see Thomae's function.

(15) If f, g are Continuous functions on $[a,b]$. and

$$\int_a^b f(x)\,dx = \int_a^b g(x)\,dx.$$

Prove that there is $c \in [a,b]$

s.t $\quad f(c) = g(c).$

Solution:

Let $h(x) = f(x) - g(x)$,

$\forall x \in [a,b]$

then 'h' is Continuous Function

Assume that $h(x) \neq 0$ for

any $x \in [a,b]$, then

either $h(x) > 0$ (or)

$h(x) < 0$ for all $x \in [a,b]$

(By Intermediate value theorem)

If $h(x) > 0$ $\exists$ $\beta > 0$ s.t

$h(x) \geq \beta$ and so,

$$\int_a^b h(x)\,dx \geq \beta(b-a) > 0$$

$$\Rightarrow \int_a^b f(x)\,dx > \int_a^b g(x)\,dx$$

$\Rightarrow\Leftarrow$

This is a contradiction.

Similarly, $h(x) < 0$ will lead contradication

$\therefore \exists c \in [a,b]$ s.t

$h(c) = 0$

$\Rightarrow f(c) = g(c).$